

Lecture 14

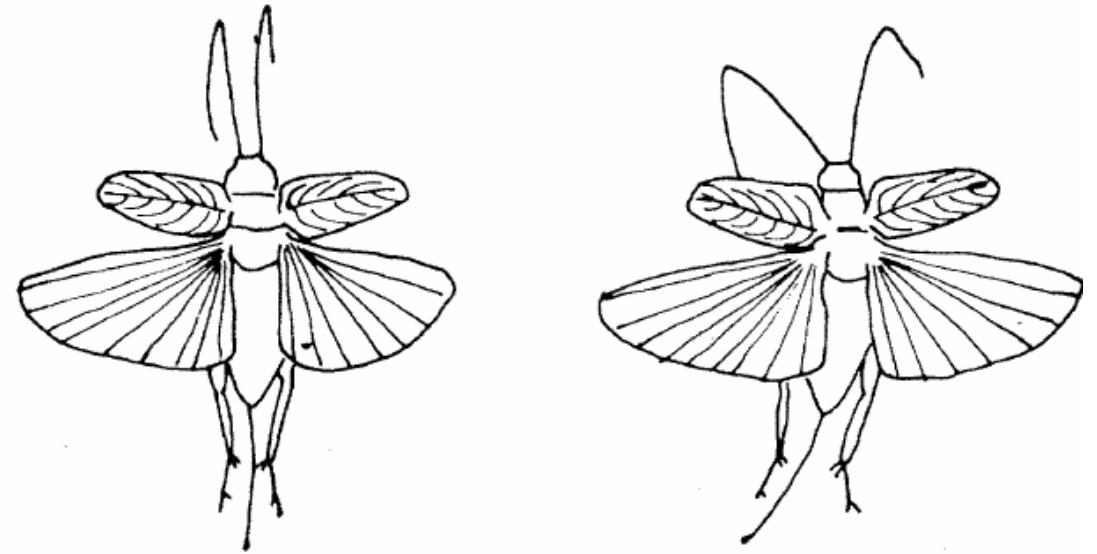
Insect hearing I: cricket phonotaxis and song recognition (Hoy)

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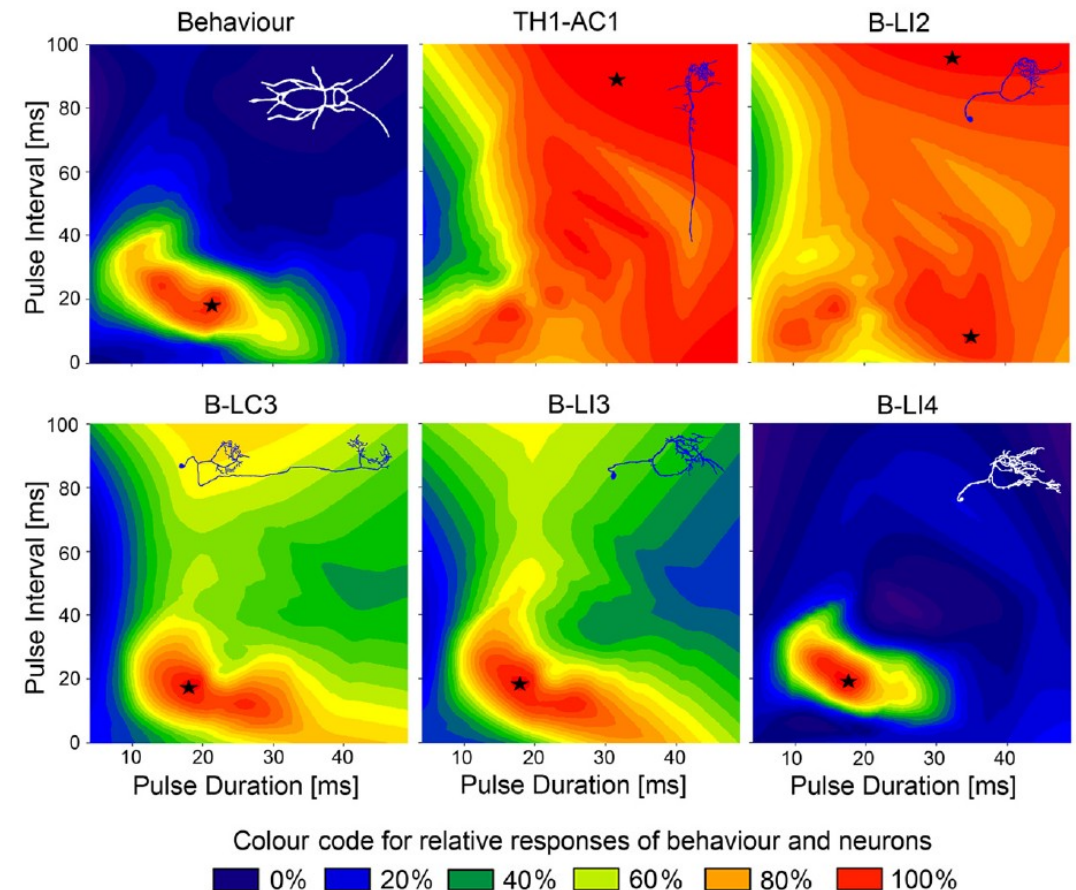
Pollack & Hoy (1979), Fig. 2(D). Tethered field cricket during flight.

Females approach species-specific calling songs

Phonotaxis is movement oriented toward or away from a sound source. Female *Gryllus bimaculatus* approach the male calling song, a repeated sequence of brief sound pulses. A **chirp** groups several pulses before a longer silent interval.

Kostarakos and Hedwig tested tethered females walking on a trackball with synthetic songs whose carrier frequency was 4.8 kHz. The **carrier frequency** specifies the rapid sound oscillation within each pulse; pulse timing describes the slower pattern of its amplitude.

Females responded most strongly to pulse periods near 34–42 ms. Controlled synthetic patterns separate frequency from timing, supporting temporal selectivity rather than indiscriminate attraction to any audible sound. Trackball steering measures orientation under tethering rather than successful arrival at a male.



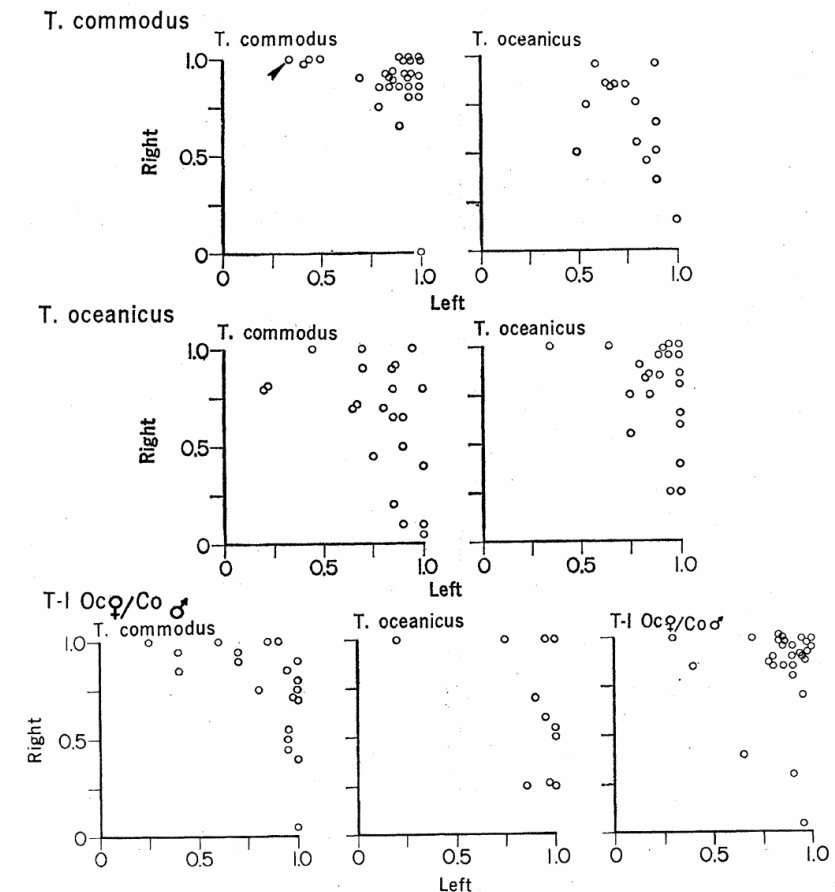
Kostarakos & Hedwig (2012), Fig. 6. Published behavior and neuronal response heat maps.

Hoy quantified directional song preference

Hoy and Paul studied *Teleogryllus oceanicus* and *T. commodus*, species with different calling-song rhythms. A **phonomotor response** was the female's tendency to turn toward a song while walking on a small rotating Y-maze rather than traveling through an arena.

Speakers were positioned 94 cm from the maze and 40° to either side. Calls were presented at 69 dB, and the active speaker changed sides. Playback and recording temperatures were matched within 2°C because cricket song timing depends on temperature.

Females preferentially turned toward their own species' song, although heterospecific songs could also attract them. Reversing speaker position distinguishes song preference from a persistent turning bias. These behavioral choices establish discrimination but do not locate its neurons or identify the genes responsible.



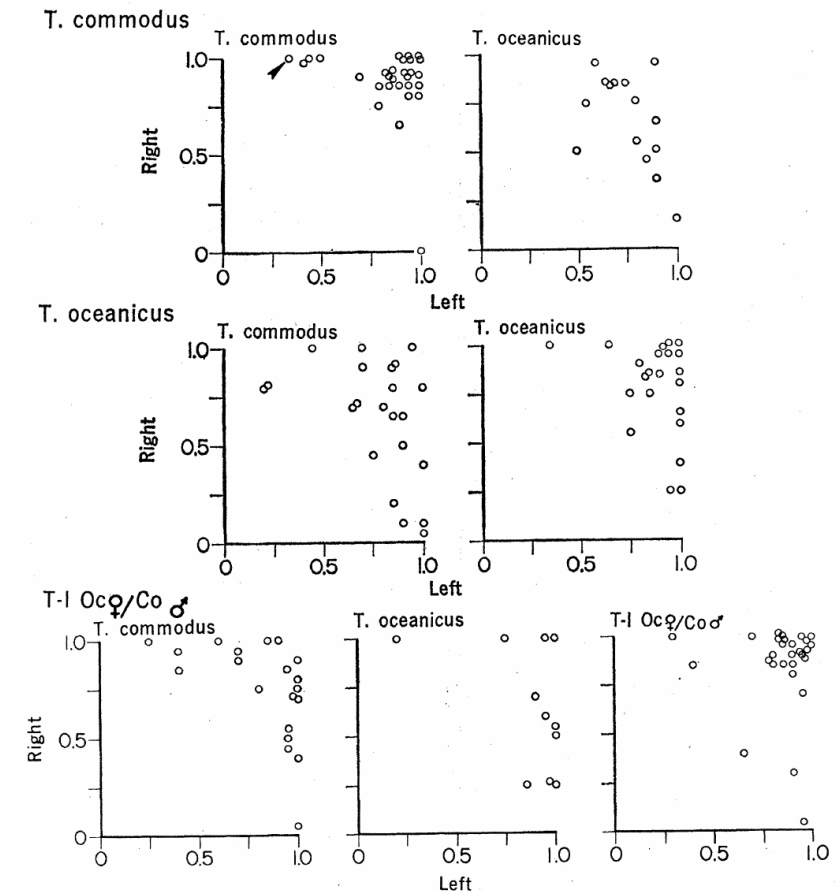
Hoy & Paul (1973), Fig. 1. Species and hybrid song preferences.

Hybrid preference supports inherited signal matching

Hybrids are offspring produced by crossing the two cricket species. Hoy and Paul compared hybrid females' responses to parental and hybrid calling songs. Hybrid male song contained rhythmic elements intermediate between the parental patterns rather than an unchanged copy of either parent's song.

The same Y-maze assay and 69-dB playback allowed parental and hybrid responses to be compared under similar acoustic conditions. Hybrid females preferred hybrid calls over either parental call; strong preference required directional responding from both speaker positions.

The result supports inherited contributions to both signal production and reception. It does not establish that the same gene, ion channel or neural oscillator causes both traits. Intermediate song and preference can arise from several genetic influences, and the experiment did not resolve their molecular identity.



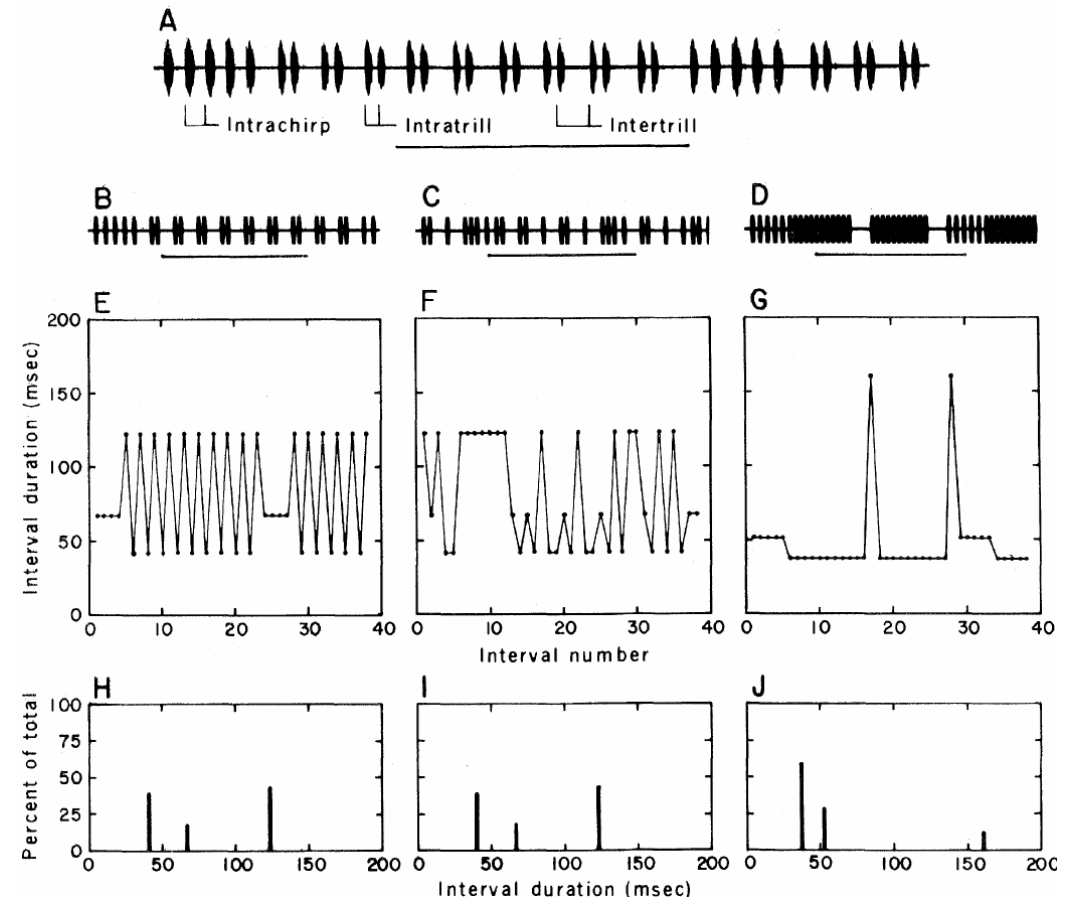
Hoy & Paul (1973), Fig. 1. Species and hybrid song preferences.

Temporal recognition can be isolated from frequency

Pollack and Hoy synthesized *Teleogryllus* songs electronically to isolate temporal structure. A **pulse period** runs from the onset of one pulse to the onset of the next, whereas a **pulse interval** ordinarily refers to the intervening silence in later cricket studies.

Their models of *T. oceanicus* and *T. commodus* song were presented with identical carrier frequencies of 4.5 or 5 kHz. Sound levels were matched within each assay: 65 dB for walking choices and 75 dB for flight steering.

Females preferred their conspecific temporal pattern in both preparations. Thus, frequency differences between the natural species' songs were unnecessary for that discrimination. The measured preference concerns the tested synthetic patterns and does not imply that frequency is irrelevant to auditory sensitivity or natural mate choice.



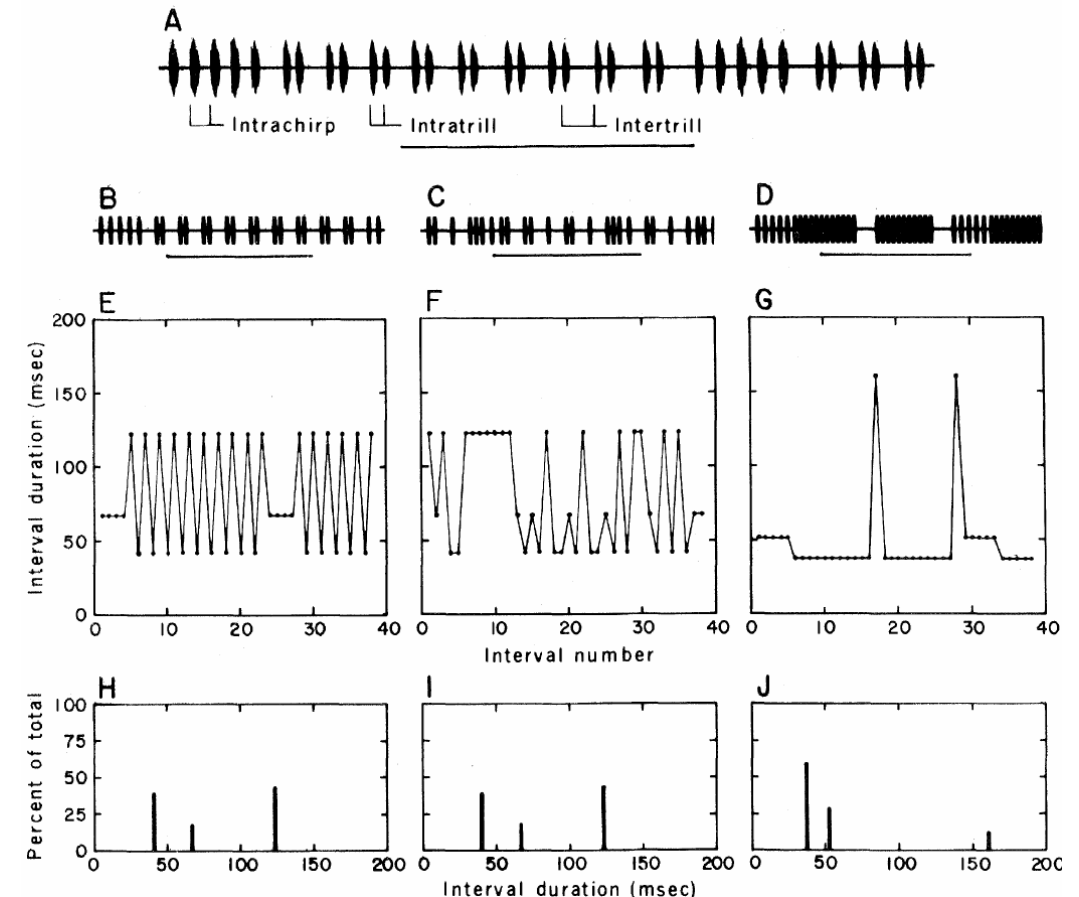
Pollack & Hoy (1979), Fig. 1(A-J). Normal, shuffled and heterospecific song intervals.

Shuffling preserves intervals but destroys sequence

A **shuffled song** retains the durations and relative numbers of the normal song's onset-to-onset intervals while rearranging their order. Pollack and Hoy used a computer to shuffle the interval sequence repeatedly and synthesize the corresponding sound pulses.

The normal *T. oceanicus* song included intervals within chirps, within trills and between trills. Its ordered progression disappeared in the shuffled stimulus, but the interval-duration distribution remained. Normal and shuffled models were played at the same 4.5- or 5-kHz carrier.

Females did not consistently discriminate normal from shuffled song in either walking or flight. Recognition therefore need not require an exact ordered copy of the complete normal phrase. The experiment leaves open shorter temporal filters and shared elements between the neural mechanisms of song production and reception.



Pollack & Hoy (1979), Fig. 1(A-J). Normal, shuffled and heterospecific song intervals.

Assay context changes apparent song discrimination

Pollack and Hoy compared a two-speaker walking-choice assay with tethered flight in a wind stream. In flight, movements of the abdomen and hind legs provided a motor correlate of attempted turning, recorded optically while speaker assignments reversed every 30 s.

Normal *T. oceanicus* song was preferred to *T. commodus* song in both assays. Shuffled song was preferred to *T. commodus* song during flight, but the walking assay did not reveal the same discrimination, despite matched song frequencies within each test.

Different motor preparations can yield different expressions of acoustic preference. The authors suggested that flight might detect smaller differences in attractiveness, but did not establish the cause of the discrepancy. Failure to discriminate in one assay is therefore insufficient to conclude that the nervous system cannot distinguish the stimuli.

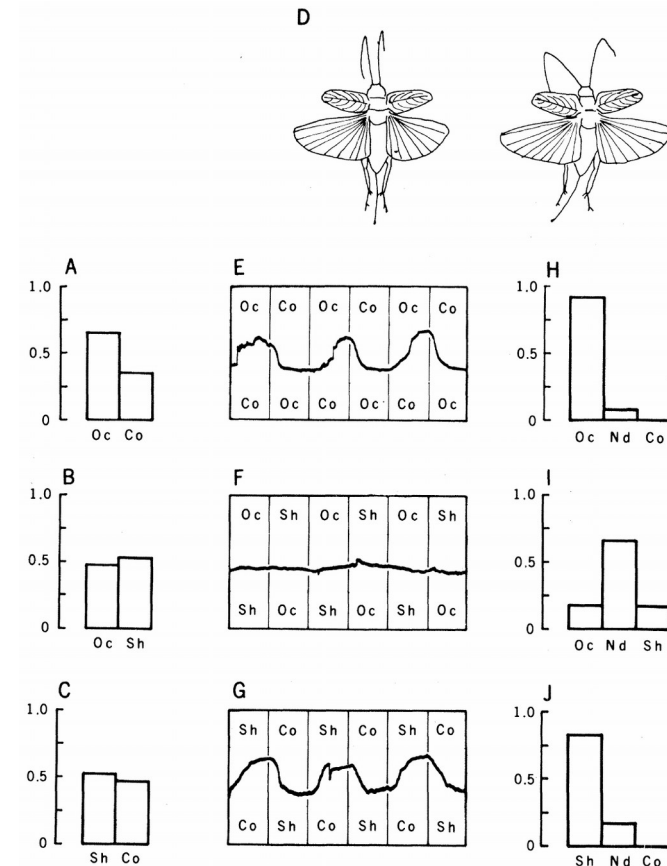


Fig. 2. (A–C) Results obtained with the walking assay for discrimination between (A) *T. oceanicus* (Oc) and *T. commodus* (Co) songs, (B) *T. oceanicus* and shuffled (Sh) songs, and (C) *T. commodus* and shuffled (Sh) songs. (D–J) Results obtained with the flight assay for discrimination between (D) *T. oceanicus* (Oc) and *T. commodus* (Co) songs, (E) *T. oceanicus* and shuffled (Sh) songs, and (F–J) *T. commodus* and shuffled (Sh) songs.

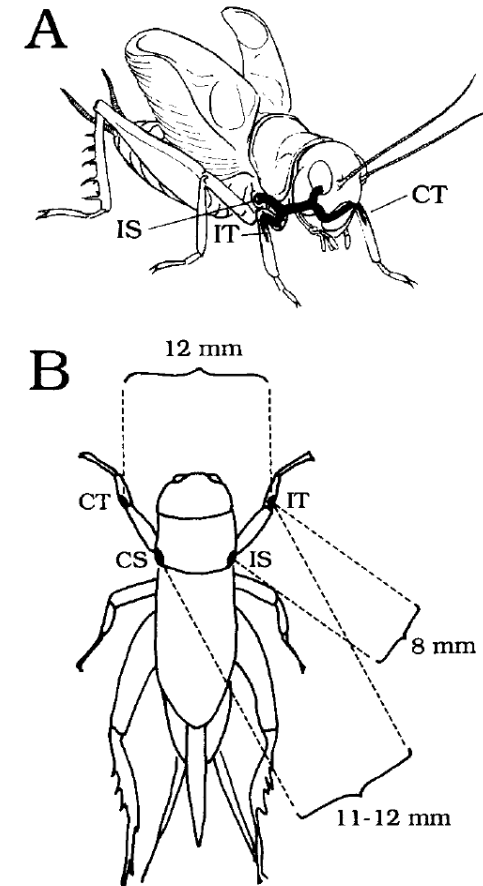
Pollack & Hoy (1979), Fig. 2(A–J). Walking choices and flight steering.

Foreleg ears receive sound through several routes

A cricket's **tympanum** is a vibrating membrane in the foreleg tibia. A **spiracle** is an opening of the tracheal air system; the thoracic acoustic spiracles provide additional routes by which sound reaches the internal surface of each tympanum.

Michelsen and colleagues measured *Gryllus bimaculatus* ears with laser vibrometry and probe microphones. The original anatomical preparation identifies tympana and spiracles separated by distances of approximately 8–12 mm, far smaller than the wavelength of the calling-song sound.

The ear operates as a **pressure-difference receiver**: vibration depends on the difference between sound acting on its outer and inner surfaces. Connected air passages permit directional interference before auditory spikes are generated. Their presence alone does not establish useful directionality; transmission amplitude and phase must also be measured.



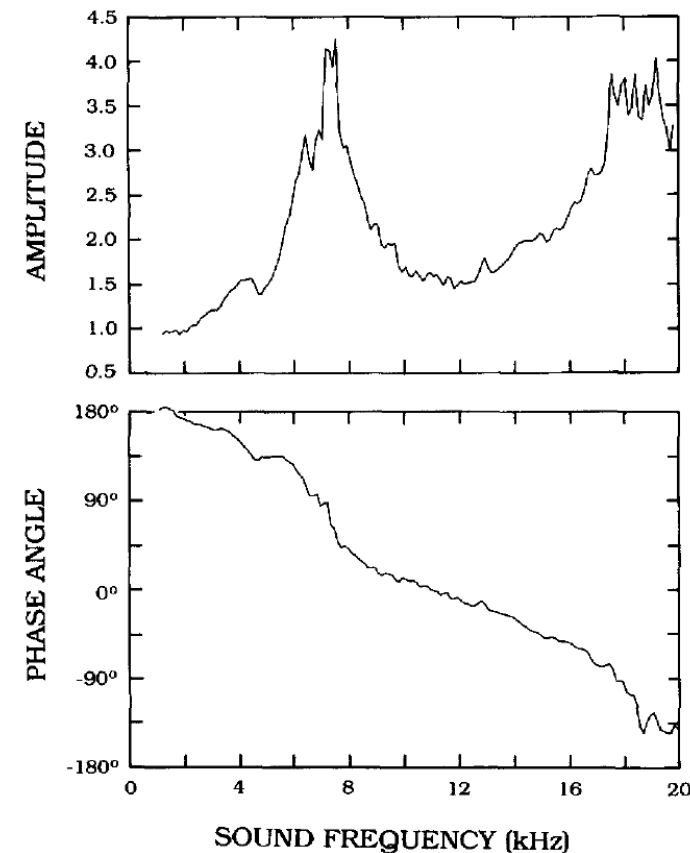
Michelsen et al. (1994), Fig. 1(A–B). Foreleg ears and acoustic inputs.

Tracheal transmission changes sound amplitude and phase

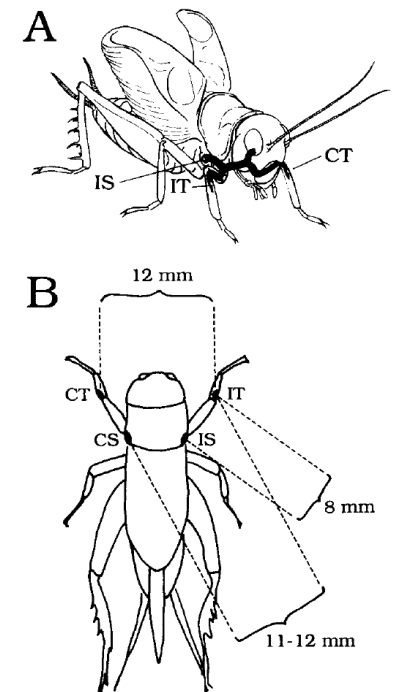
Transmission gain describes how strongly a sound component reaches the tympanum through an internal route. **Phase** describes its position within the sound's oscillatory cycle. Two components can reinforce or oppose one another depending on both quantities.

Michelsen and colleagues applied local sound at one acoustic input while measuring tympanal vibration. Wax barriers reduced sound at other inputs by at least 20 dB. Ipsilateral-spiracle transmission gain peaked near 4–5 kHz, the calling-song frequency range.

The measured gain and phase specify how an internal sound component combines with the external input. A large internal amplitude does not automatically imply better localization because reinforcement depends on timing. The isolated-input preparation characterizes acoustic transmission rather than intact walking behavior.



Michelsen et al. (1994), Fig. 5. Ipsilateral spiracle transmission gain and phase.



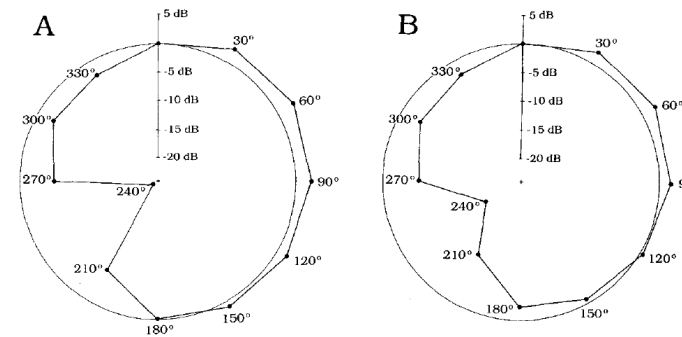
Michelsen et al. (1994), Fig. 1(A–B). Foreleg ears and acoustic inputs.

Opposite spiracle input sharpens directionality

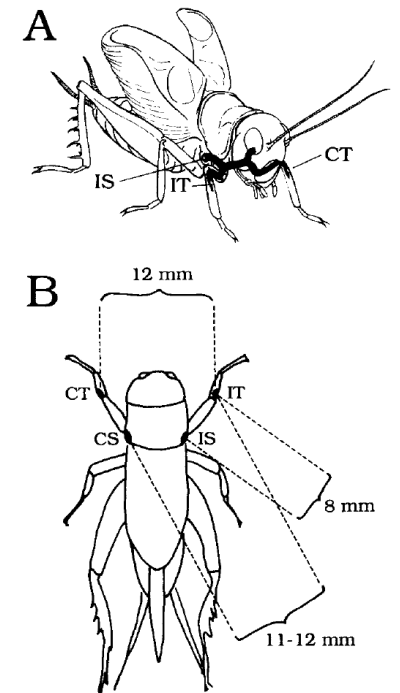
For sound arriving from ahead, left and right external inputs differ only slightly because the cricket's body is small. The contralateral acoustic spiracle supplies an internal component whose phase can convert that modest spatial difference into directional tympanal vibration.

Michelsen and colleagues calculated directional responses from measured input gains and phases. The opposite-spiracle component changed phase by approximately $130\text{--}140^\circ$ as sound direction moved from 0° to 30° off the midline, strongly changing the predicted interference at the ear.

Including both spiracle routes produced forward directional tuning; omitting the contralateral contribution altered that pattern substantially. This supports a mechanical role for cross-body sound transmission. The calculation uses measured acoustics, but it does not identify a behavioral decision rule or establish how central neurons weight the resulting ear signals.



Michelsen et al. (1994), Fig. 11(A–B). Contralateral input changes directional tuning.



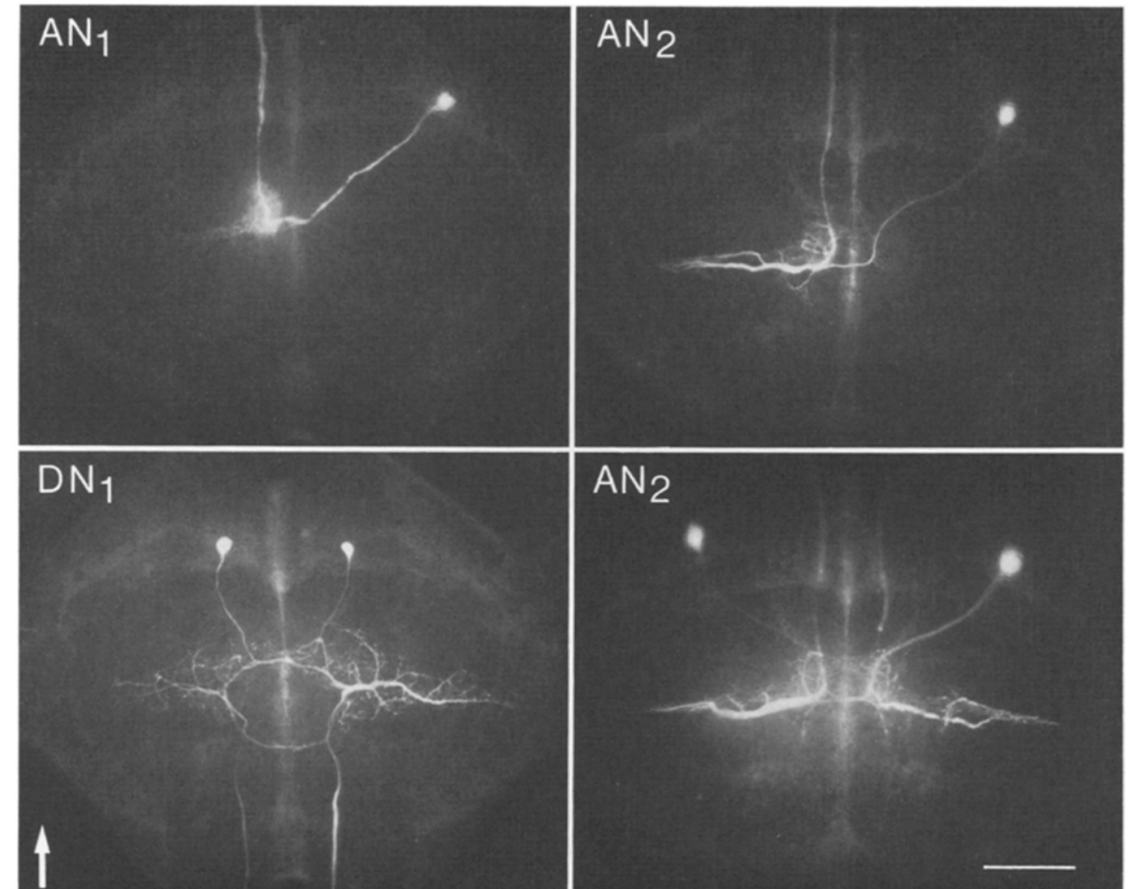
Michelsen et al. (1994), Fig. 1(A–B). Foreleg ears and acoustic inputs.

Thoracic auditory cells occupy distinct anatomical routes

The **prothoracic ganglion** is the segmental neural center associated with the first thoracic segment and forelegs. Auditory sensory axons terminate in its auditory **neuropil**, a region containing interacting neuronal branches rather than a collection of cell bodies alone.

Wohlers and Huber recorded individual neurons in female *Gryllus campestris* and filled them with Lucifer dye. Stimuli covered 2–20 kHz, and filled processes were examined after recording to connect a cell's acoustic responses with its anatomical projection.

Ascending neurons carried signals toward the head, whereas local omega neurons remained within the thoracic ganglion. The fluorescent micrographs establish identified routes for early auditory processing. Dye labeling reveals morphology, but overlapping branches alone do not prove a functional synapse or its transmitter identity.



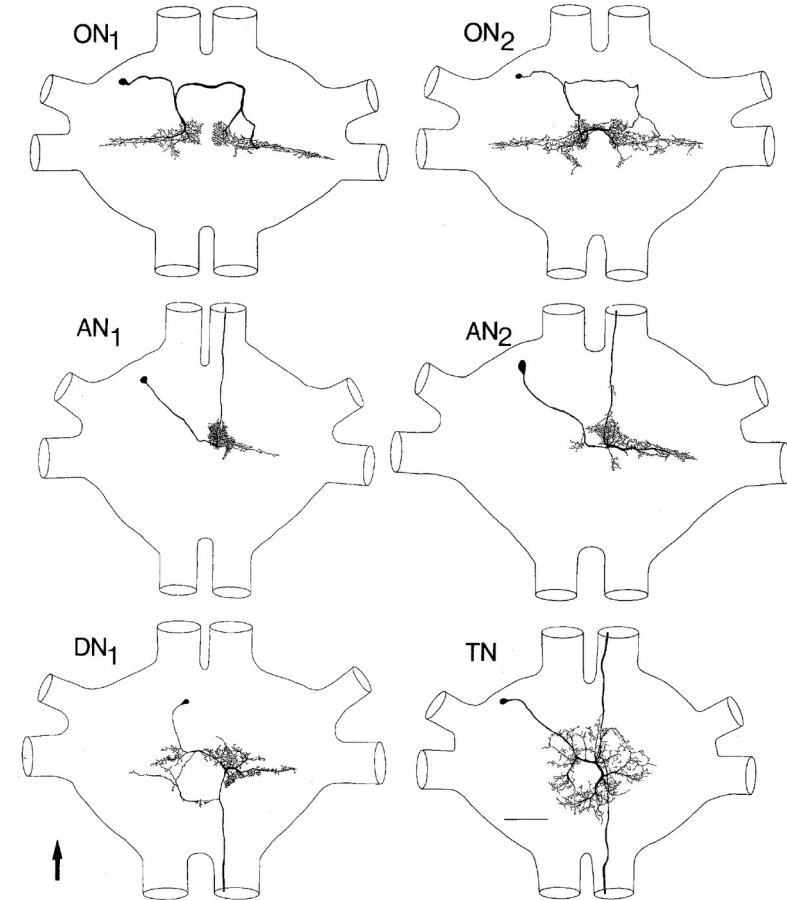
Wohlers & Huber (1982), Fig. 1. Dye-labeled auditory interneurons.

Six cell types distribute early auditory information

Wohlers and Huber identified six types of auditory interneuron arranged as bilateral mirror-image pairs. An **interneuron** receives and processes neuronal inputs rather than directly serving as a sensory receptor or the final motor neuron supplying a muscle.

ON1 and ON2 had omega-shaped local branches; AN1 and AN2 projected upward toward the brain; DN1 projected downward; and TN had both ascending and descending branches. Their responses were characterized with synthetic chirps and 2–20-kHz stimulation under separate-ear conditions.

These routes permit local comparisons and transmission of acoustic information to different neural centers. The authors' anatomical reconstructions distinguish projection patterns but do not assign each neuron a unique behavioral function. A neuron that responds to calling song is not necessarily the cell that recognizes its species-specific rhythm.



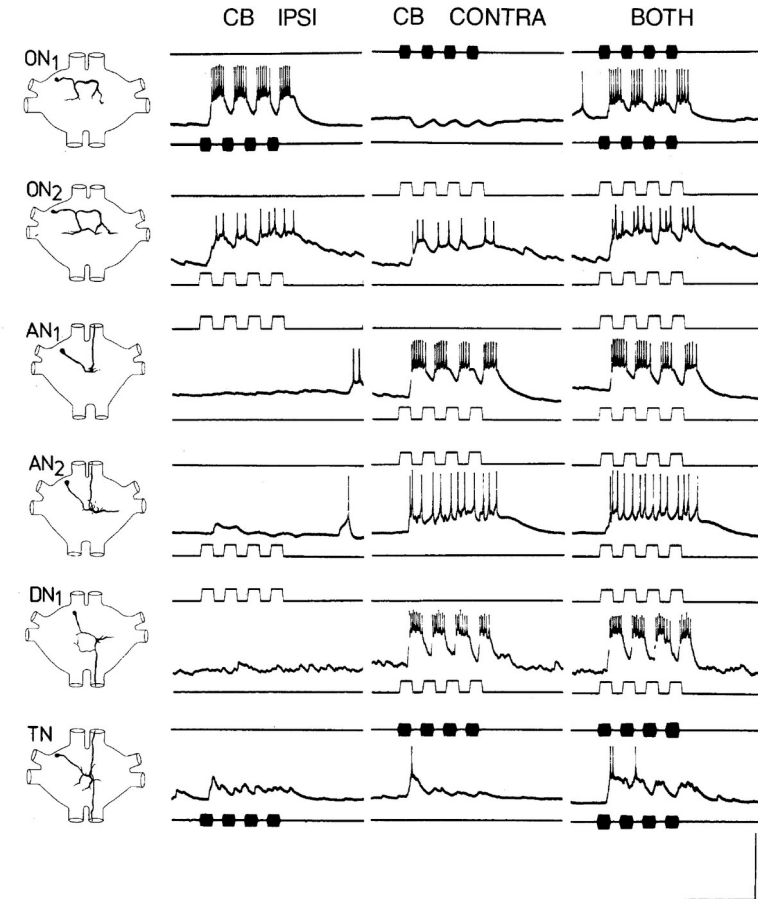
Wohlers & Huber (1982), Fig. 2. Six identified neuron types.

ON1 combines excitation with opposite inhibition

Excitation increases a neuron's tendency to fire, whereas **inhibition** reduces that tendency through synaptic effects on its membrane potential or conductance. In ON1, the ear on the cell-body side supplies excitation and the opposite ear supplies inhibition.

Wohlers and Huber separated the ears acoustically with legphones, cut the connecting trachea and sealed thoracic spiracles. Four-pulse calling-song stimuli at 4.5 kHz were presented to each ear alone and then together while recording intracellularly from identified cells.

ON1's response to bilateral stimulation reflects interacting neural inputs rather than acoustic cross-talk alone. This configuration supports an early neural comparison between sides. It establishes the signs of the recorded responses, but does not by itself determine which other cell supplies the inhibition or which receptor mediates it.



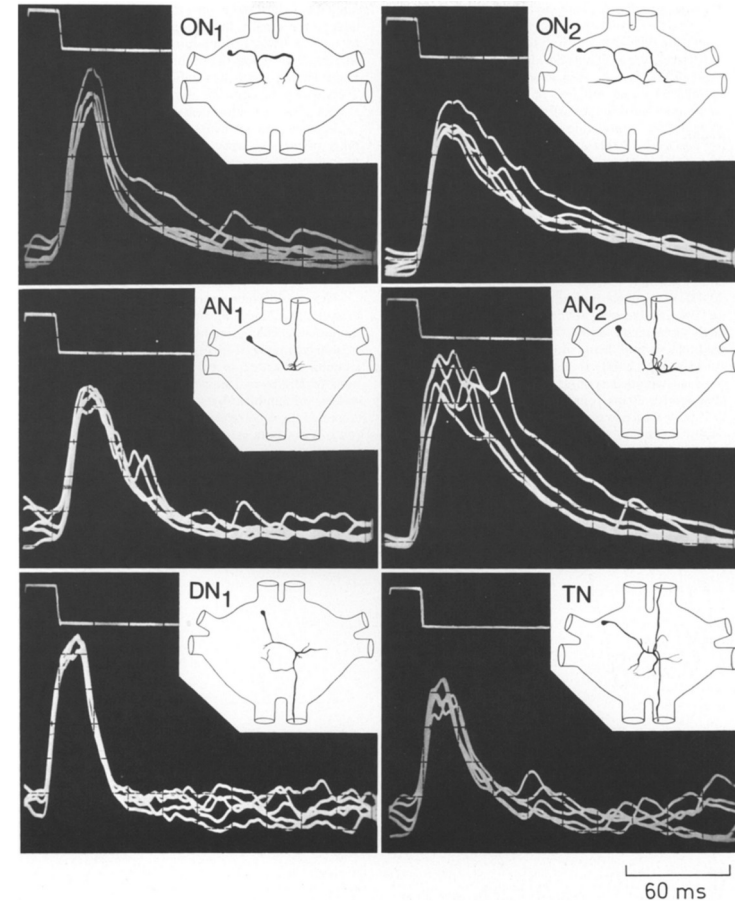
Wohlers & Huber (1982), Fig. 6. Separate-ear and bilateral stimulation.

AN1 copies pulses without recognizing the pattern

AN1, the first ascending auditory neuron, carries calling-song information from thoracic neuropil toward the brain. Its excitatory response is driven predominantly by the ear opposite its cell body, a laterality convention that differs from the excitatory side of ON1.

Wohlers and Huber recorded responses to 1-syllable chirps across sound frequencies. A 20-ms pulse near 4.5 kHz produced excitation, and AN1 followed the pulse-and-pause structure of calling-song patterns more accurately than several other thoracic neurons.

Accurate copying makes AN1 a useful input to later temporal processing, but copying is distinct from selectivity. The thoracic cell continued responding to patterns outside the behaviorally preferred interval range. Its spikes represent available sound events without establishing that the cricket has accepted those events as a conspecific song.



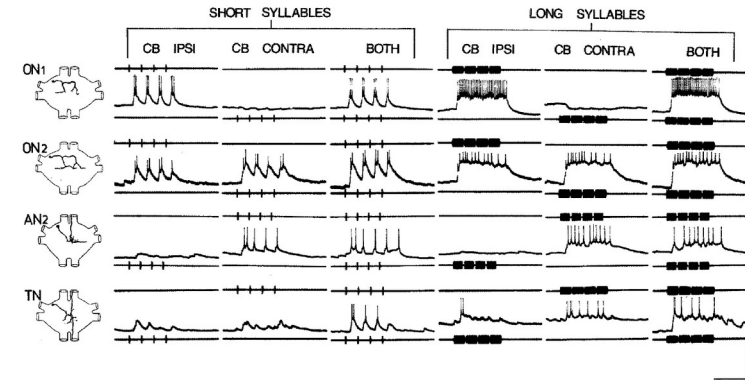
Wohlers & Huber (1982), Fig. 4. Intracellular responses and neuron shapes.

Thoracic resolution exceeds recognition selectivity

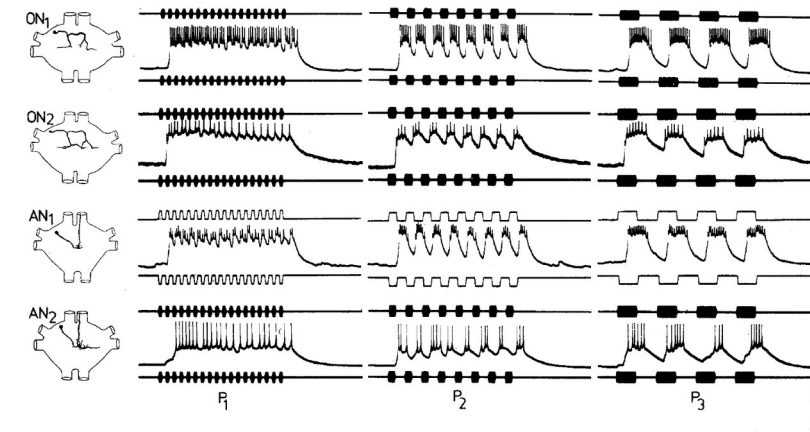
Temporal resolution is the ability of a response to preserve separate sound events. It differs from temporal preference, which requires a stronger response to particular timing values. A neuron may resolve many pulse patterns without favoring the species-specific pattern.

Wohlers and Huber tested 2.5-ms syllables and progressively shorter pauses. The six thoracic neuron types could copy those brief pulses, but none reliably copied pauses shorter than 5 ms. Equal-duration syllable and pause patterns were also tested over different repetition periods.

Behavior was preferentially elicited by approximately 20–60-ms silent intervals in the compared conditions, yet the recorded thoracic cells were not specifically tuned to that entire range. The difference places a recognition problem downstream of simple event copying. It does not exclude contributions from thoracic timing or adaptation to the later computation.



Wohlers & Huber (1982), Fig. 8. Responses to short and long syllables.



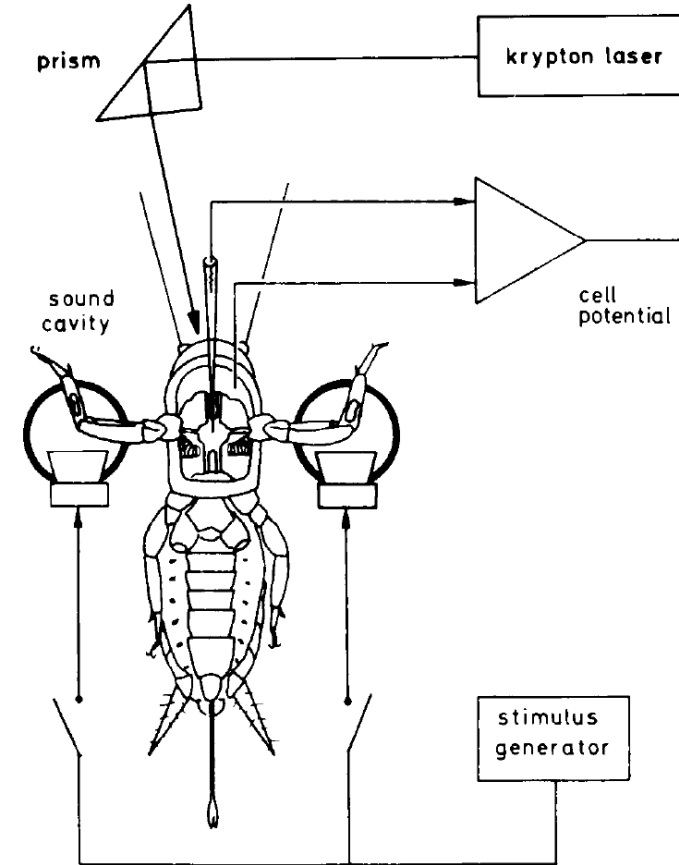
Wohlers & Huber (1982), Fig. 9. Responses to varied syllable periods.

Photoinactivation tests a cell's causal role

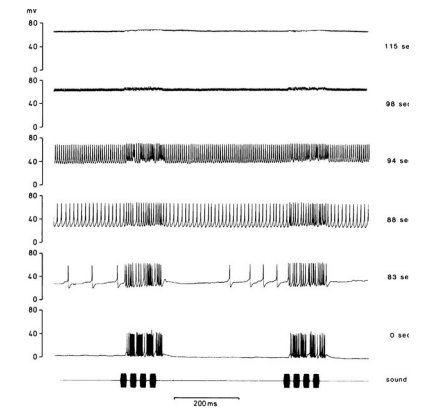
Photoinactivation selectively damages a dye-filled neuron by illuminating the preparation with intense light. Selverston and colleagues filled an identified auditory interneuron with Lucifer Yellow, then tested whether its loss changed the responses of another cell.

Female *Gryllus bimaculatus* received separate-ear, 5-kHz chirps through legphones. Laser irradiation at 5–50 mW progressively depolarized the stained neuron and abolished its activity. The acoustic preparation reduced cross-talk between ears by as much as 60 dB.

Comparing responses before and after loss provides causal evidence beyond anatomical overlap. A changed postsynaptic response can identify a required source of inhibition, provided the preparation remains functional. The method removes a cell's influence rather than selectively blocking one transmitter receptor or distinguishing every synapse that cell makes.



Selverston et al. (1985), Fig. 1. Legphones and selective photoinactivation.



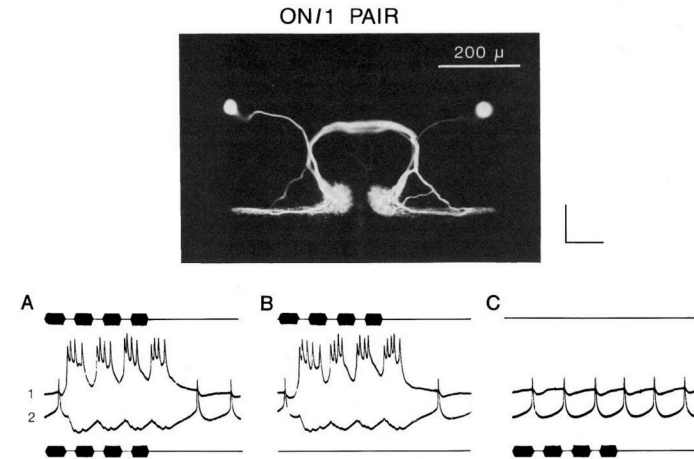
Selverston et al. (1985), Fig. 2. Loss of activity during irradiation.

Omega cells provide reciprocal inhibition

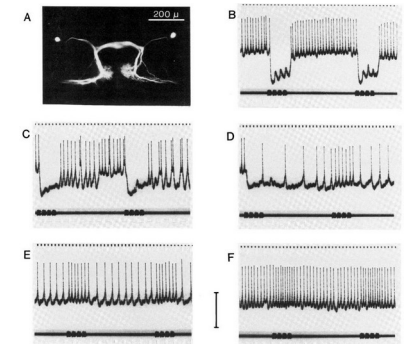
Reciprocal inhibition means that each member of a neuronal pair inhibits the other. Selverston and colleagues combined simultaneous intracellular recording from paired ON1 cells with selective photoinactivation to test that organization directly.

Separate-ear, 5-kHz stimulation excited the corresponding ON1 and suppressed activity in its partner. After one stained ON1 was inactivated, the surviving cell lost its strong opposite-ear inhibitory response and a weak excitatory response became apparent.

The paired recordings and ablation support ON1 as a source of reciprocal suppression, sharpening the contrast between competing auditory inputs. The unmasked excitation means the opposite-ear pathway is not purely inhibitory. These recordings do not establish the chemical transmitter or show that ON1 alone is sufficient for phonotactic steering.



Selverston et al. (1985), Fig. 4. Paired ON1 recordings.



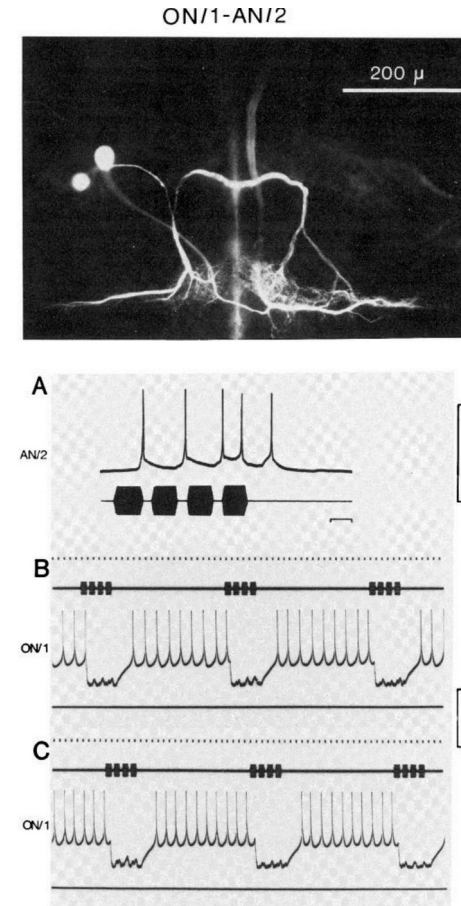
Selverston et al. (1985), Fig. 7(A–F). ON1 ablation removes opposite-ear inhibition.

ON1 also supplies inhibition to an ascending cell

AN2 is an ascending auditory interneuron distinct from the calling-song pathway AN1. In the photoinactivation experiments, its response combined strong excitation from one ear with inhibition from stimulation on the cell-body side.

Selverston and colleagues recorded AN2 during separate-ear, 5-kHz chirps before and after damaging the ipsilateral ON1. The previously inhibitory response was replaced by weak excitation, while the neuron's other acoustic responses could still be recorded.

This intervention identifies ON1 as a source of AN2 inhibition and supports a network in which local thoracic cells regulate ascending output. It does not justify assigning the same connectivity to every ascending cell. In particular, an experimentally demonstrated ON1–AN2 interaction must not be relabeled as an ON1–AN1 synapse.



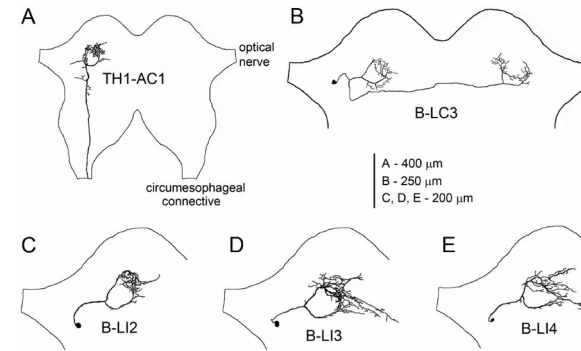
Selverston et al. (1985), Fig. 11(A–C). ON1 ablation removes AN2 inhibition.

Brain neurons branch near AN1 terminals

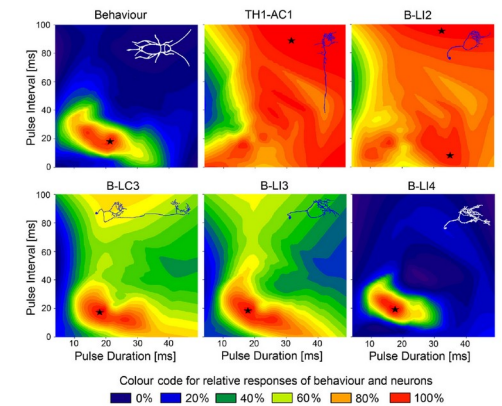
The **anterior protocerebrum** is a front region of the cricket brain containing terminals of the ascending auditory pathway. Kostarakos and Hedwig identified local brain cells whose ring-like branches overlapped that termination area in female *Gryllus bimaculatus*.

Intracellular recordings were followed by dye labeling while synthetic 4.8-kHz chirps varied in duration, interval and period. Their labels B-LI2, B-LI3 and B-LI4 correspond to LN2, LN3 and LN4 in the later circuit study; B-LC3 connected the auditory regions across the midline.

Different response selectivities occurred among cells occupying the same local region. The anatomy supports a compact substrate for processing temporal patterns, while the recorded tuning distinguishes information copying from recognition-related filtering. Branch overlap is evidence of proximity, not direct proof of every proposed synaptic connection.



Kostarakos & Hedwig (2012), Fig. 1(A–E).
Ascending and local brain neurons.



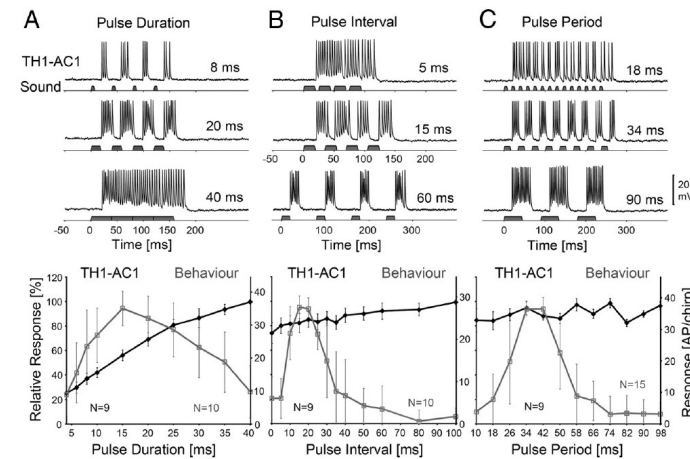
Kostarakos & Hedwig (2012), Fig. 6.
Published behavior and neuronal
response heat maps.

AN1 spike count does not encode attractiveness

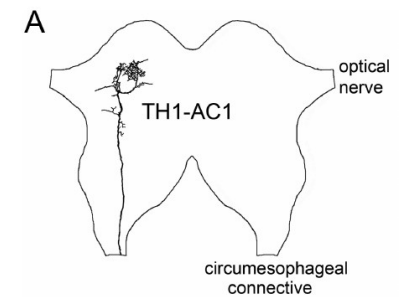
An **action potential** is a brief regenerative electrical spike that can transmit activity along an axon. Kostarakos and Hedwig counted AN1 action potentials per chirp and compared them with phonotactic steering elicited by the same temporal stimulus classes.

Increasing pulse duration from 8 to 20 to 40 ms increased AN1 output from approximately 14 to 26 to 37 spikes per chirp. Behavioral attraction instead peaked near 15-ms pulses and decreased toward longer durations under this stimulus paradigm.

The divergence means that total ascending spike count is insufficient to explain female preference. Later processing must distinguish the temporal organization carried by AN1. The comparison supports a transformation of the input representation, but it does not alone identify the necessary synapses or exclude additional ascending contributions under other acoustic conditions.



Kostarakos & Hedwig (2012), Fig. 2(A–C). AN1 activity and behavioral tuning.



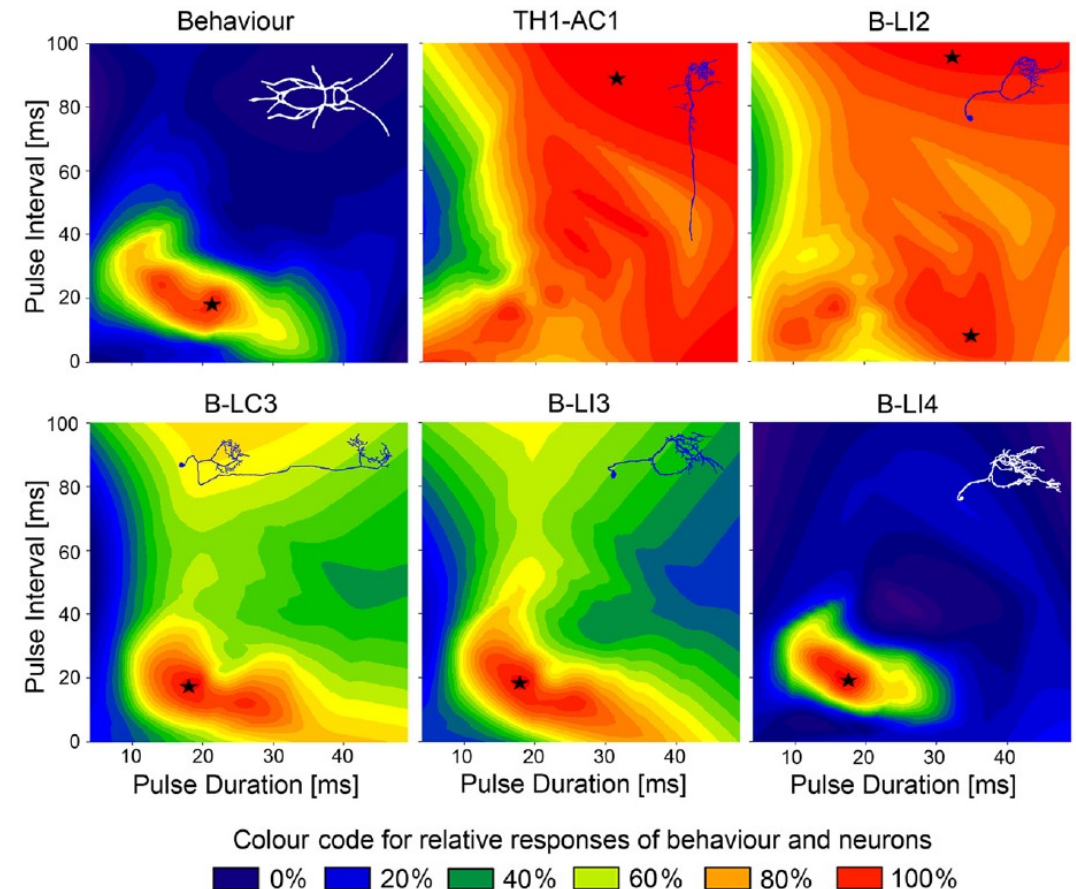
Kostarakos & Hedwig (2012), Fig. 1(A). AN1 projection (400 μ m common scale).

Brain neurons filter pulse duration

Pulse duration specifies how long sound remains on during each pulse. Kostarakos and Hedwig varied this duration while maintaining a 40-ms pulse period, so longer pulses necessarily reduced the silent interval and changed both aspects of the temporal pattern.

At 4.8 kHz, early local neuron B-LI2 continued to copy many patterns, whereas B-LI4 exhibited a more selective response that broadly matched phonotaxis. B-LI4 received inhibition as well as excitation, and unattractive patterns could produce strong inhibition rather than reliable spiking.

The findings place temporal selectivity within brain processing rather than in AN1's total activity. Because duration and silence covaried, the test does not isolate a pure duration detector. Its interpretation requires comparison with interval and period paradigms rather than assigning all selectivity to a single stimulus variable.



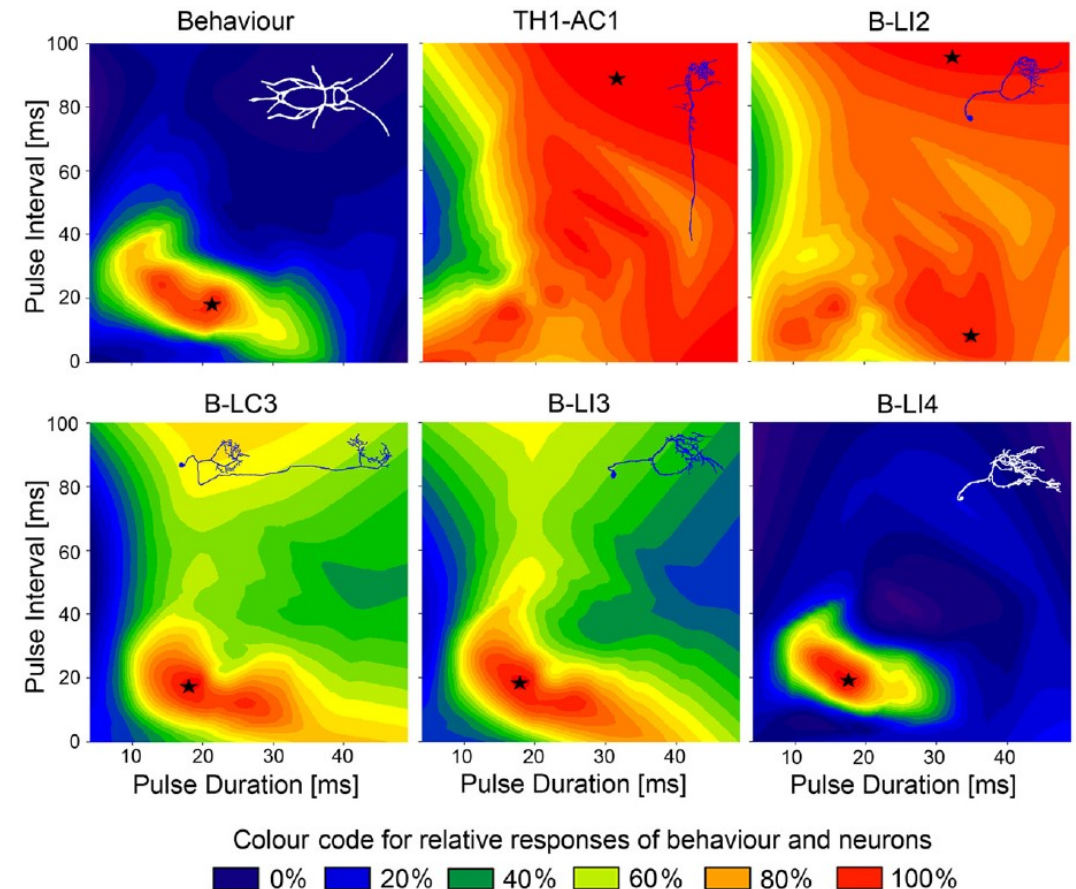
Kostarakos & Hedwig (2012), Fig. 6. Published behavior and neuronal response heat maps.

Paired pulses reveal interval selectivity

A **temporal band-pass response** favors an intermediate range of timing values and weakens at both shorter and longer values. Kostarakos and Hedwig tested local brain cells with 20-ms pulses separated by different silent intervals.

B-LI4 responded particularly strongly when the second pulse followed a species-appropriate interval near 20 ms. Other local neurons responded to a wider range of timings, including patterns that produced little phonotaxis. The stimulus carrier remained 4.8 kHz.

Selectivity can therefore emerge by the second pulse rather than requiring an entire long song phrase. The interaction of excitation and inhibition during the interval is consistent with a short temporal computation. These recordings identify response properties but do not, by themselves, establish the full delay-generating mechanism or its molecular components.



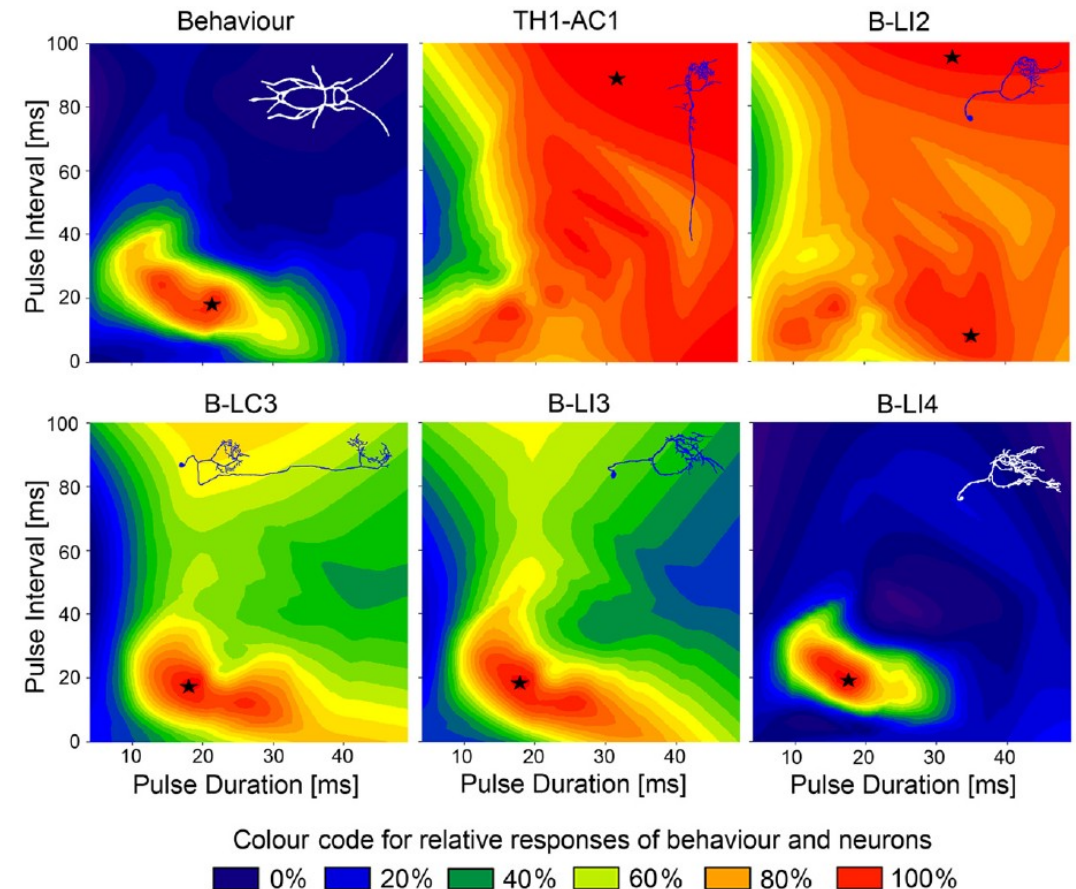
Kostarakos & Hedwig (2012), Fig. 6. Published behavior and neuronal response heat maps.

Constant-energy tests reveal period selectivity

Pulse period combines sound duration and the following silent interval. Kostarakos and Hedwig used a constant-sound-energy paradigm in which pulse number and duration were adjusted to avoid making a more attractive timing pattern simply deliver more total sound.

Female phonotaxis was strongest around 34–42 ms and declined toward shorter and longer periods. AN1's spike output did not reproduce this band-pass profile, whereas the local B-LI4 response more closely matched behavioral selectivity across the tested periods.

Controlling acoustic energy strengthens the interpretation that temporal structure contributes to attraction. The method does not hold every individual temporal feature constant, and matched neural and behavioral tuning remains correlational. It supports a recognition-related output stage while leaving the causal relationship to descending steering pathways unresolved.



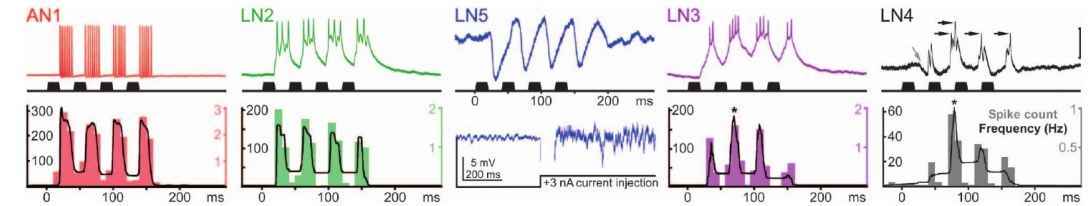
Kostarakos & Hedwig (2012), Fig. 6. Published behavior and neuronal response heat maps.

The second pulse recruits a different brain response

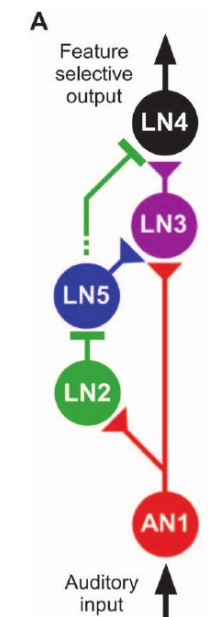
Schöneich and colleagues recorded the ascending neuron AN1 and four local brain cells during standard 4.8-kHz chirps. The chirps contained four 20-ms sound pulses separated by 20-ms silent intervals, providing repeated opportunities for interaction between successive responses.

AN1 and LN2 produced their strongest responses to the first pulse and adapted somewhat over the chirp. LN3 and LN4 responded most strongly to the second pulse; LN4 often failed to spike to the first despite receiving synaptic input.

The contrasting sequences support a transformation of sensory copying into timing-dependent output. A **nonspiking neuron**, LN5, changed membrane voltage without generating action potentials even during depolarizing current injection. Its graded response provides a distinct candidate mechanism for carrying temporally delayed information within the local brain circuit.



Schöneich et al. (2015), Fig. 2. Responses across the five-neuron auditory pathway.



Schöneich et al. (2015), Fig. 5(A). Proposed auditory circuit.

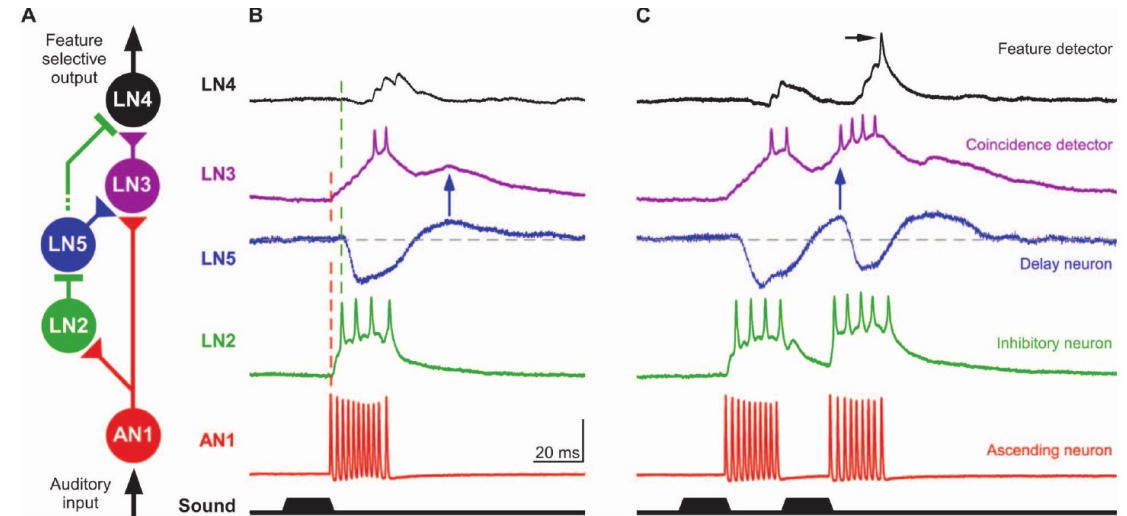
AN1 supplies fast excitation in the brain

In the proposed recognition circuit, AN1 supplies a direct excitatory pathway to local neurons LN2 and LN3. An **EPSP** is a depolarizing postsynaptic response; successive EPSPs can sum until a neuron crosses its spike threshold.

At 75 dB SPL, AN1's brain response began about 20 ms after sound onset. LN2's EPSPs began at approximately the same latency, and two large EPSPs were sufficient to trigger its spikes. LN3 required summation of roughly five or six smaller EPSPs.

Different integration requirements allow the same ascending pulse representation to produce different output timing. The inference of direct excitation rests on response latencies, frequency tuning and EPSP structure.

Simultaneous paired intracellular recordings were not feasible, so each proposed connection should not be described as independently proven by a paired synaptic experiment.



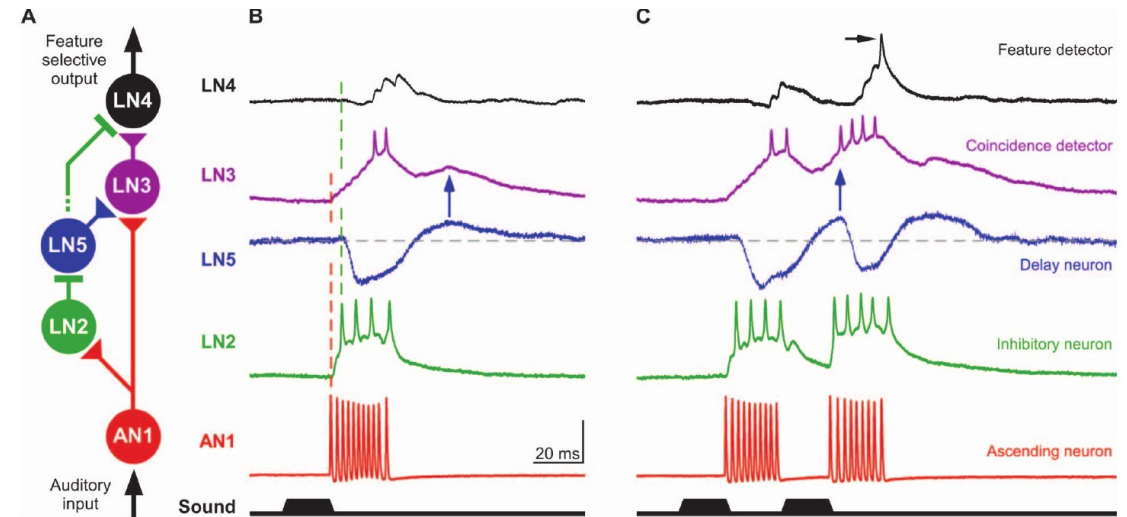
Schöneich et al. (2015), Fig. 5(A–C). Proposed connections and measured response timing.

LN2 provides a rapid inhibitory branch

LN2 is a local spiking brain interneuron whose activity closely follows the ascending acoustic input. Its response precedes inhibition in both the nonspiking delay neuron LN5 and the selective output neuron LN4 in the proposed network.

LN2 spiked approximately 23 ms after sound onset, while LN4 and LN5 began inhibitory responses near 25 ms. Above about 50 dB SPL, LN2 output was relatively insensitive to further increases in sound intensity, unlike AN1's more intensity-dependent activity.

This timing and shared intensity dependence support LN2 as the inhibitory branch linking acoustic input to the two downstream cells. Inhibitory processing is established physiologically, but the study did not chemically identify its transmitter or receptor. Naming a particular GABA receptor would therefore exceed the evidence provided by these recordings.



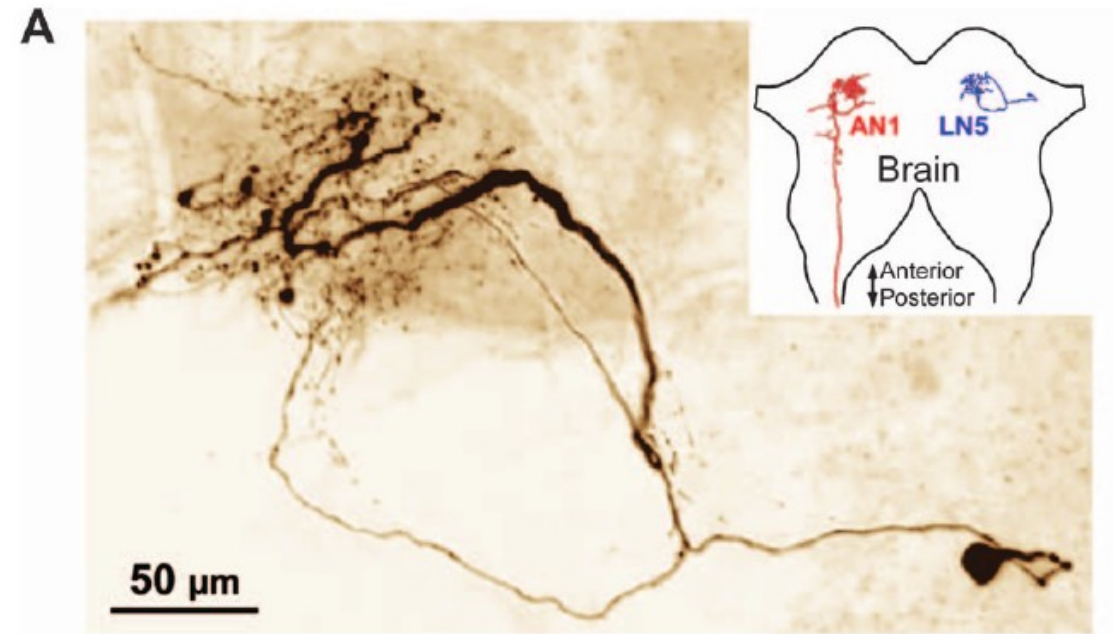
Schöneich et al. (2015), Fig. 5(A–C). Proposed connections and measured response timing.

LN5 is an identified nonspiking delay-line neuron

LN5 has a ventrolateral cell body and branches near AN1's terminals in the anterior protocerebrum. Its anatomical location places it within the local auditory network rather than in the thoracic sensory ganglion or a long ascending tract.

Schöneich and colleagues recorded sound-evoked hyperpolarization followed by depolarization in LN5. Standard 20-ms pulses produced inhibition of approximately 4–9 mV and a later depolarization of 3–5 mV above rest. Depolarizing current injection did not evoke spikes.

These properties distinguish a graded local signal from an axonal spike train. The delayed membrane depolarization can contribute excitation to later circuit processing, but morphology and voltage recording do not identify its transmitter. The published confocal image preserves the cell's labeled arborization and 50- μ m scale rather than reconstructing a new anatomical diagram.



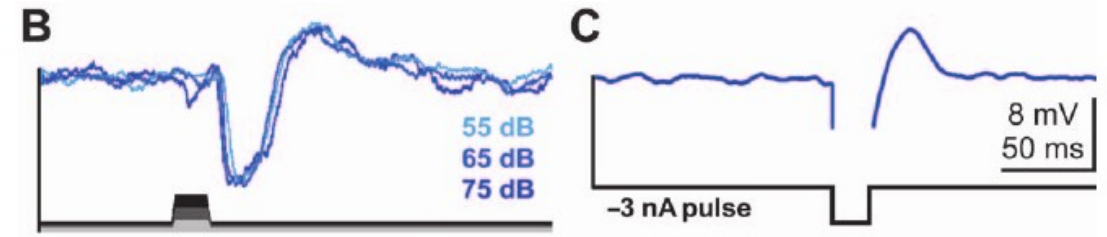
Schöneich et al. (2015), Fig. 4(A). LN5 anatomy (50 μ m).

Release from inhibition generates rebound

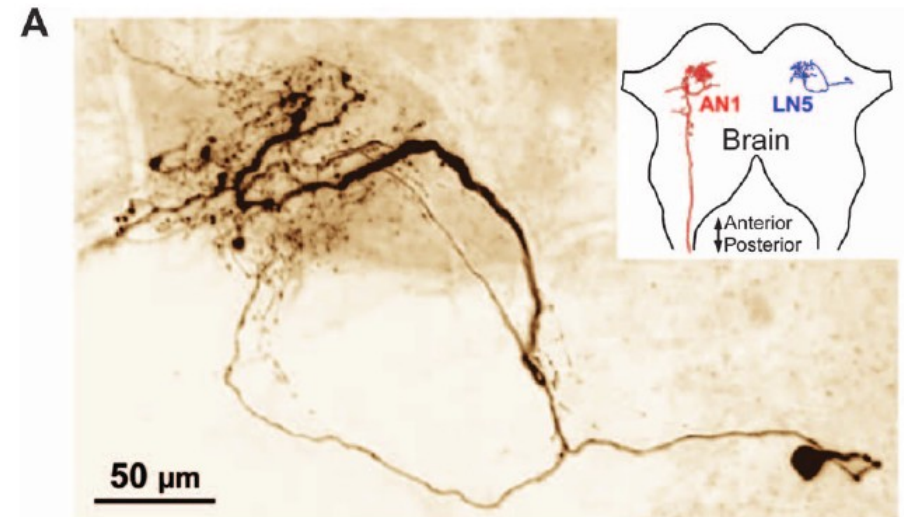
Postinhibitory rebound is a depolarization occurring after inhibition ends. A delayed excitatory synaptic input could produce a superficially similar waveform, so Schöneich and colleagues tested whether LN5 could rebound when the auditory stimulus was replaced by injected current.

A 20-ms hyperpolarizing current pulse of -3 nA produced a 3–5-mV depolarization after release. Sound pulses at 55, 65 and 75 dB SPL also elicited similar rebound responses despite differences in stimulus intensity.

The current-injection control supports an intrinsic membrane contribution to the delayed signal rather than requiring delayed acoustic excitation alone. It does not identify the responsible conductance or prove that the rebound has a single molecular source. The cell's voltage response establishes a physiological mechanism more directly than its transmitter or ion-channel identity.



Schöneich et al. (2015), Fig. 4(B–C). Sound and injected current evoke rebound.



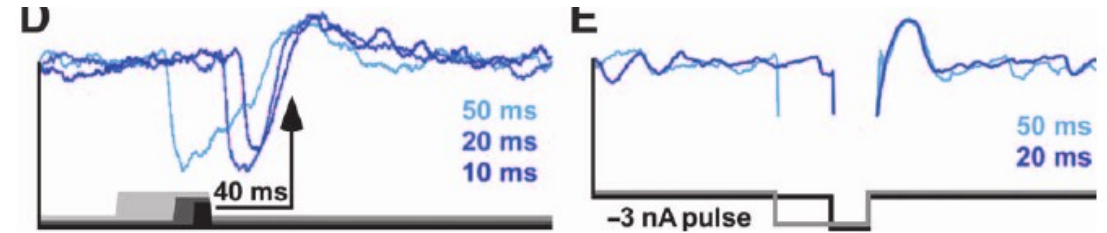
Schöneich et al. (2015), Fig. 4(A). LN5 anatomy (50 μ m).

Rebound timing follows sound offset

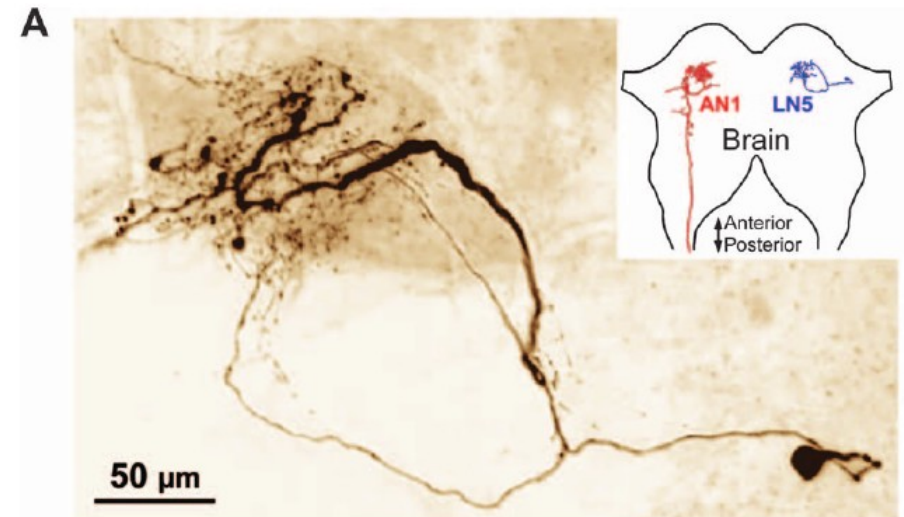
A delay may be coupled to sound onset or to sound offset, and these alternatives predict different effects when pulse duration changes. LN5's rebound was linked to release from hyperpolarization rather than to a fixed time after the initial acoustic event.

Schöneich and colleagues varied sound-pulse and current-pulse duration. Across the tested acoustic durations of 10–50 ms, the rebound peaked approximately 43 ms after sound offset. Current-evoked rebound likewise followed the end of the imposed hyperpolarization.

The delayed signal therefore marks the transition out of inhibition and can align with excitation produced by a following pulse. This is a cellular timing operation, not merely conduction through a longer axon. Its timing was measured under the experimental conditions; temperature or species differences were not systematically mapped in this experiment.



Schöneich et al. (2015), Fig. 4(D–E). Rebound follows release from inhibition.



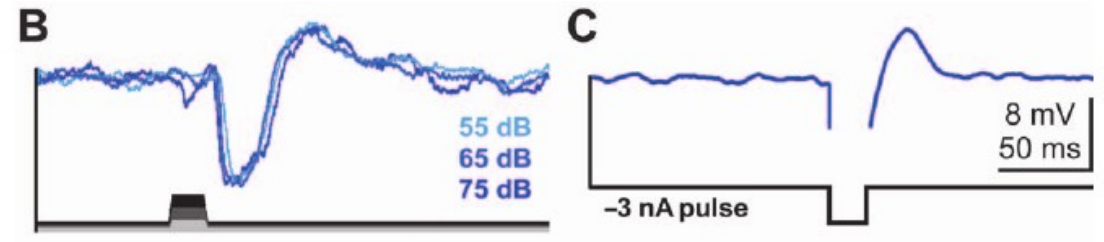
Schöneich et al. (2015), Fig. 4(A). LN5 anatomy (50 μm).

The rebound channel remains unidentified

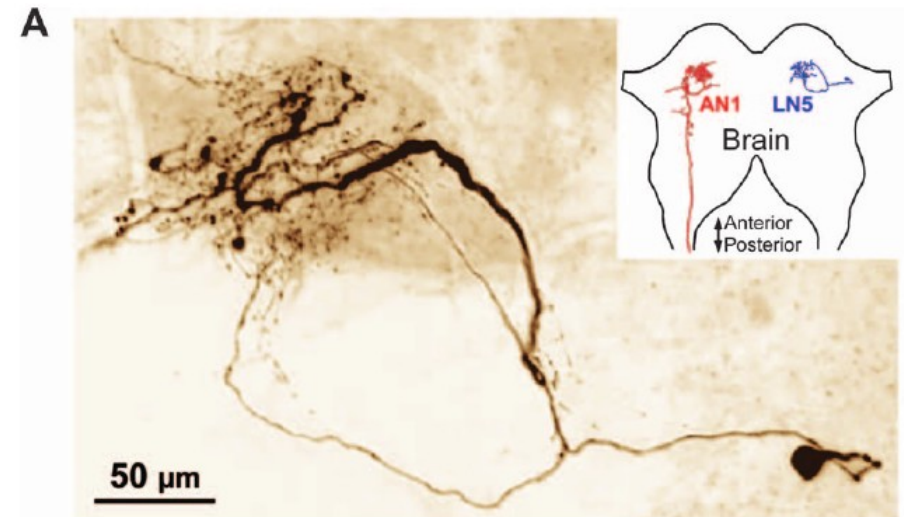
A **membrane conductance** is the ease with which ions cross the neuronal membrane through channels. A **hyperpolarization-activated channel** opens more strongly at negative membrane voltages and could, in principle, contribute to depolarization during or after inhibition.

LN5's 3–5-mV rebound after a -3 -nA current pulse supports an intrinsic electrical mechanism. Schöneich and colleagues proposed that shared control of membrane proteins might couple timing in sender and receiver circuits, but did not pharmacologically isolate a channel in LN5.

The measured sequence is inhibition, release, rebound and delayed excitation. A specific channel subtype, transmitter or receptor cannot be inserted into that sequence as an established finding. The hypothesis links genetic matching to cellular timing plausibly, while the actual protein identity and its causal contribution to cricket song recognition remain unresolved by the study.



Schöneich et al. (2015), Fig. 4(B–C). Sound and injected current evoke rebound.



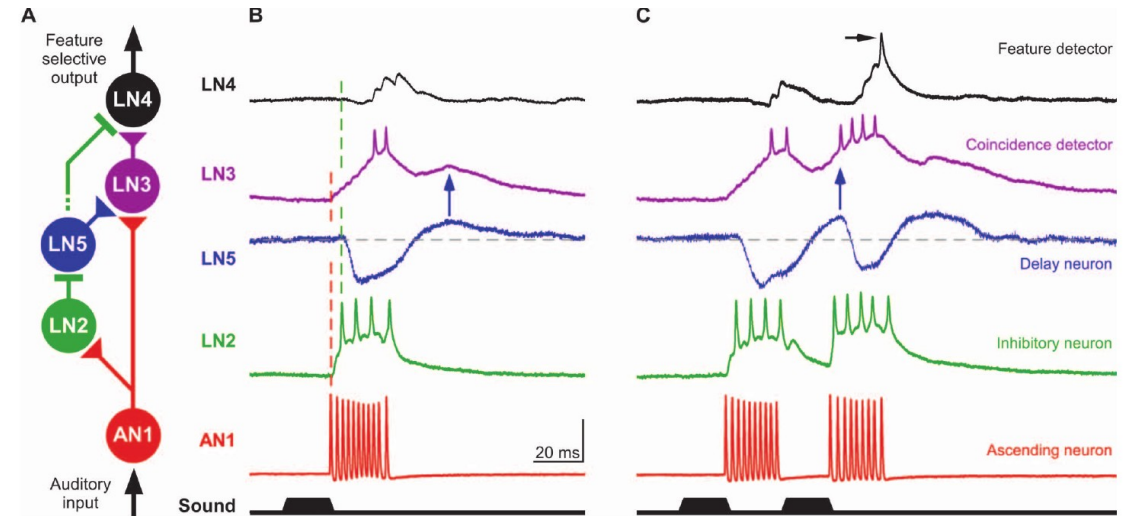
Schöneich et al. (2015), Fig. 4(A). LN5 anatomy (50 μ m).

LN3 combines direct input with delayed excitation

A **coincidence detector** responds more strongly when separate inputs arrive together. In the proposed cricket circuit, LN3 receives rapid excitation associated with AN1 spikes and delayed excitation associated with LN5's rebound after the previous sound pulse.

For two 20-ms pulses separated by a 20-ms silent interval, the direct response to the second pulse coincides with delayed depolarization following the first. LN3's second-pulse spike response increases, and its spike latency falls from approximately 34 ms for the first pulse to 29 ms for the second.

The timing supports coincidence detection based on synaptic integration rather than an exact stored copy of the complete chirp. Direct and delayed pathways were inferred from their response properties, not all established by simultaneous recordings. LN3's enhanced response supplies a mechanism for temporal selectivity but does not itself specify the entire motor output.



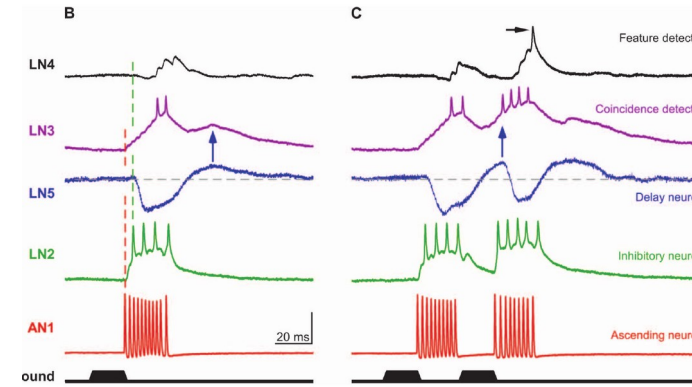
Schöneich et al. (2015), Fig. 5(A–C). Proposed connections and measured response timing.

LN3 favors an intermediate interval between pulses

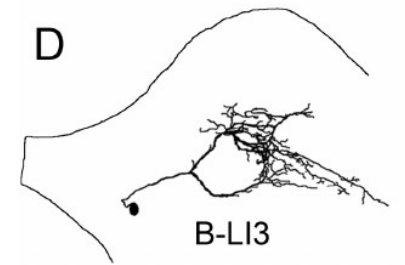
Schöneich and colleagues tested LN3 with pairs of 20-ms pulses separated by different silent intervals. A long interval allows the first delayed response to finish before the second direct response, whereas a very short interval merges or truncates the two responses.

At a 60-ms interval, LN3 generated similar responses to the two pulses. At 20 ms, its output increased from approximately 1.6 spikes to 3.2 spikes for the second pulse. The strongest enhancement occurred over intervals of about 15–25 ms.

This intermediate optimum is consistent with direct and delayed excitation summing during an appropriate temporal window. The cell also responds outside that window, so it is less selective than the final output neuron. Its tuning should not be equated with a complete behavioral acceptance decision or an absolute inability to detect other sounds.



Schöneich et al. (2015), Fig. 5(B–C). Responses to one and two pulses.



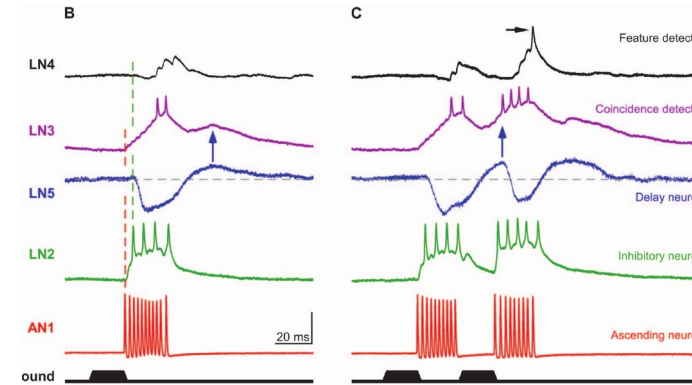
Kostarakos & Hedwig (2012), Fig. 1(D). LN3 (200 μ m common scale).

LN4 integrates excitation and inhibition

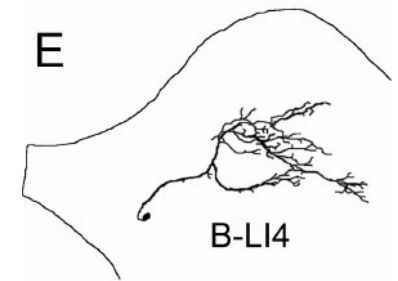
A **feature detector** responds preferentially to a stimulus property associated with a meaningful signal. LN4 combines the inhibitory branch attributed to LN2 with excitation following LN3's coincidence-enhanced activity in the proposed song-recognition circuit.

A single 20-ms pulse usually produced inhibition followed by subthreshold depolarization in LN4. A second pulse after a 20-ms interval evoked approximately 1.4 spikes. Excitatory responses in LN4 followed LN3 spikes with a relatively constant delay near 3 ms.

The second-pulse excitation can overcome inhibition when the interval is appropriate, creating a more selective output than LN3 alone. LN4 is not the primary coincidence detector because it does not receive the delayed excitatory component directly. These relationships support the proposed sequence but do not identify the transmitter or prove that LN4 alone is necessary for approach.



Schöneich et al. (2015), Fig. 5(B–C). Responses to one and two pulses.



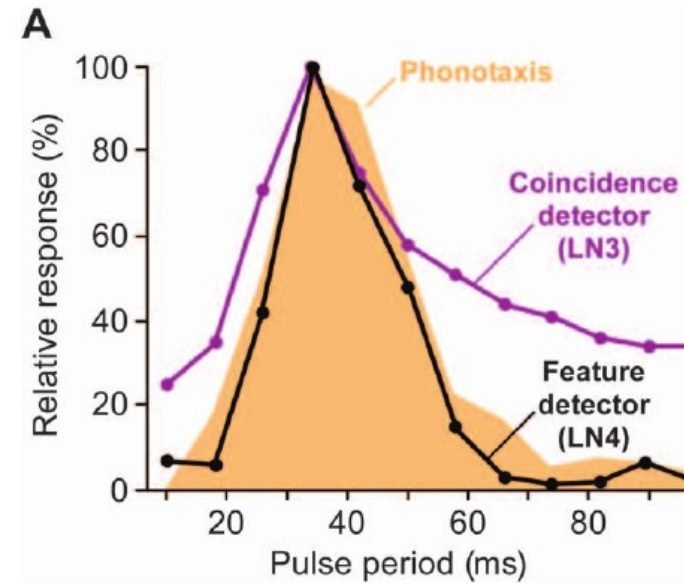
Kostarakos & Hedwig (2012), Fig. 1(E). LN4 (200 μ m common scale).

LN4 tuning matches phonotactic preference

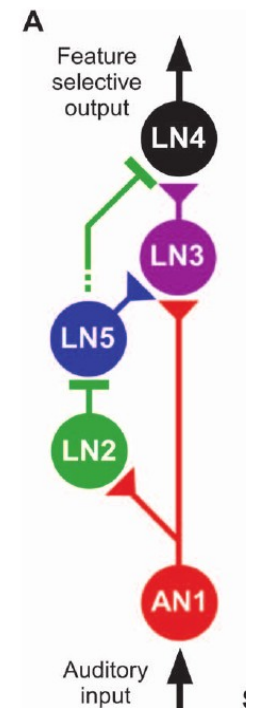
Schöneich and colleagues compared neuronal pulse-period tuning with female phonotaxis using constant-sound-energy chirps. **Band-pass selectivity** favors intermediate periods and rejects both rapidly repeated and widely separated pulses rather than merely increasing with sound duration.

Across pulse periods of 10–98 ms, behavioral attraction peaked around 34 ms. LN3 and LN4 had similar optima, but the stronger filtering in LN4 more closely matched the behavioral response. The original figure plots these separately measured curves together.

The match links a physiological mechanism to the animal's accepted timing range while avoiding a simple sound-energy explanation. It remains evidence of correspondence rather than a cell-specific behavioral intervention. Connections from the local feature detector to descending motor pathways and the necessity of that detector were not established by the tuning comparison.



Schöneich et al. (2015), Fig. 6(A). Neural and behavioral pulse-period tuning.



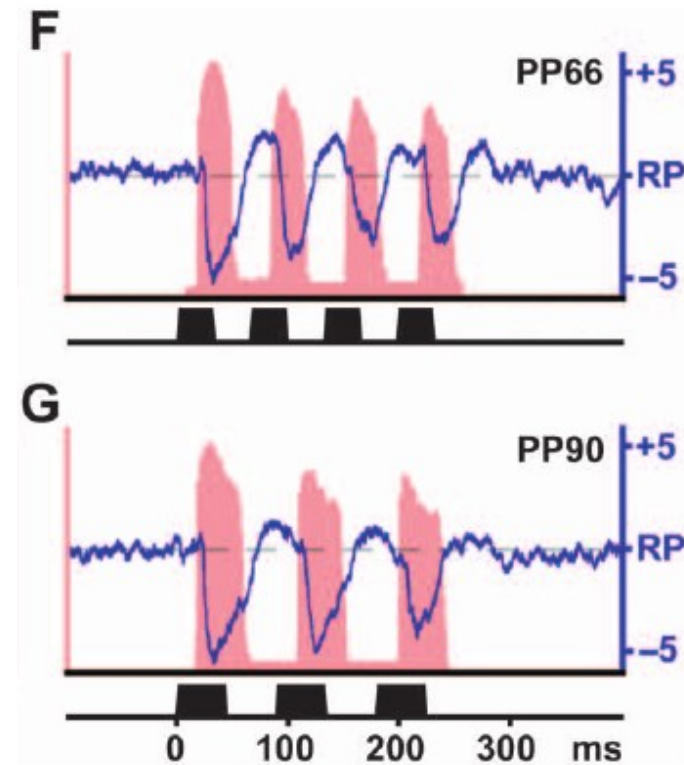
Schöneich et al. (2015), Fig. 5(A). Proposed auditory circuit.

Slow pulses separate the two excitatory pathways

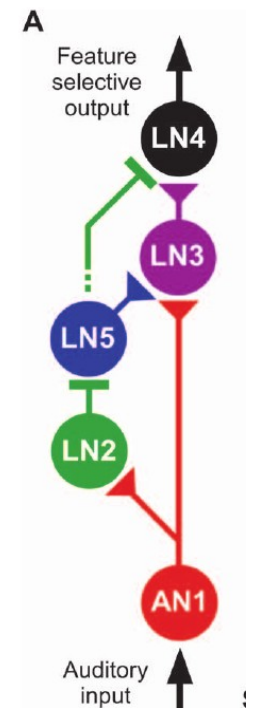
At slow pulse repetition, the delayed response to one event and the direct response to the next occur at different times. They therefore contribute less simultaneous excitation to LN3 even though AN1 can still report the individual sound pulses.

For pulse periods of 66–98 ms, Schöneich and colleagues found that AN1 spike activity and LN5 rebound depolarization drifted out of phase. These values corresponded to silent intervals of 33–49 ms in the tested constant-energy patterns.

Loss of overlap explains the circuit's reduced response to slow patterns without requiring failure of auditory detection. It establishes the slow-period side of the band-pass mechanism. The interpretation depends on measured timing in the direct and delayed pathways and should not be generalized to all cricket species with different natural song periods.



Schöneich et al. (2015), Fig. 6(F–G). Slow pulses separate direct and delayed activity.



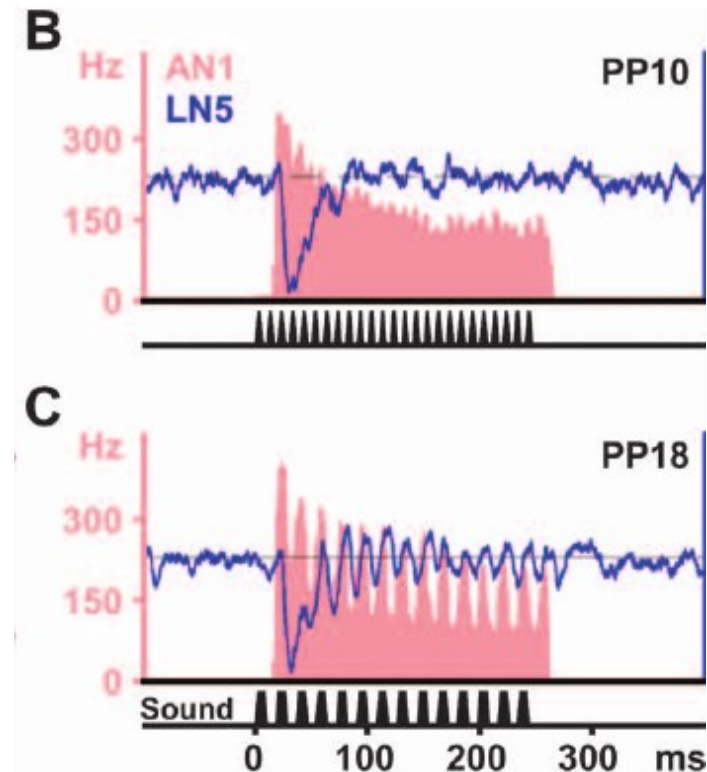
Schöneich et al. (2015), Fig. 5(A). Proposed auditory circuit.

Fast pulses truncate rebound and weaken the input

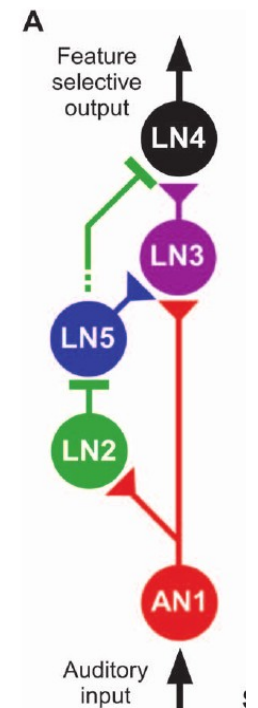
Rapid repetition does not automatically create stronger coincidence. A new pulse can recruit inhibition before the previous LN5 rebound has developed, while adaptation reduces the fidelity and strength of the ascending acoustic response.

Schöneich and colleagues tested pulse periods below 34 ms, corresponding to intervals shorter than 17 ms. LN5 rebound was increasingly truncated by following inhibition, and AN1 adapted strongly enough to lose accurate copying of the fastest tested pulse patterns.

These effects weaken both the delayed and direct contributions to LN3 and establish the fast-period side of band-pass tuning. A delay line alone would predict responses to some faster repetitions; the measured membrane and adaptation processes constrain that simpler account. Their relative necessity was not separated by selectively eliminating each process during behavior.



Schöneich et al. (2015), Fig. 6(B–C). Fast pulses truncate rebound.



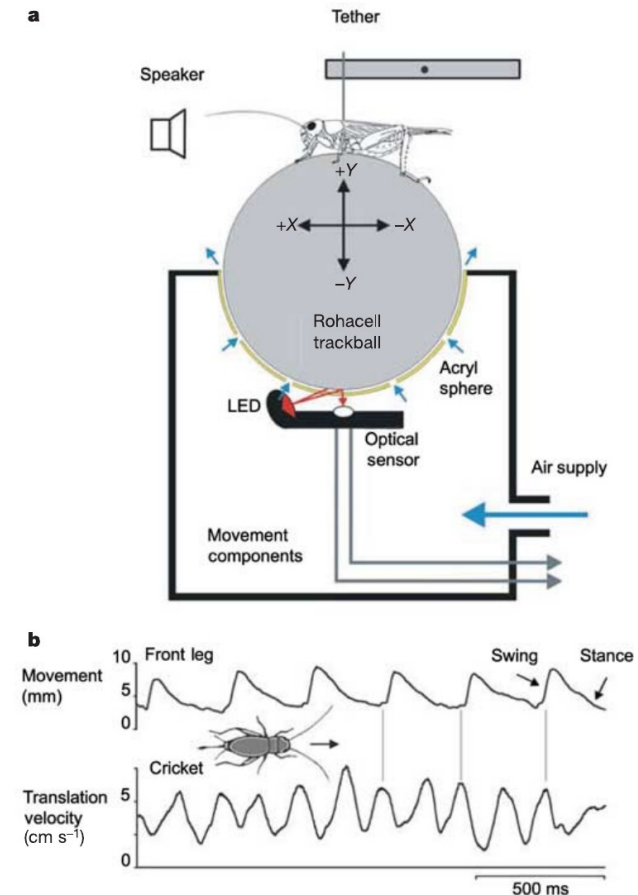
Schöneich et al. (2015), Fig. 5(A). Proposed auditory circuit.

A fast trackball resolves pulse-linked steering

Hedwig and Poulet measured female *Gryllus bimaculatus* walking on a lightweight trackball floating in air. An optical sensor registered small ball movements, allowing lateral steering to be measured separately from forward walking rather than averaged over a whole chirp.

The system had a temporal resolution of 0.3 ms. During acoustic approach, females walked at approximately 5–7 cm/s, and their lateral velocity could change rapidly in response to individual pulses. Slow trackball measurements had previously averaged over much longer time windows.

The preparation reveals motor responses that a coarse measure can merge into an apparently smooth trajectory. High temporal resolution does not create a new recognition behavior; it changes which components of the existing behavior are observable. Tethering and the absence of actual translation still limit direct extrapolation to free-field approach paths.



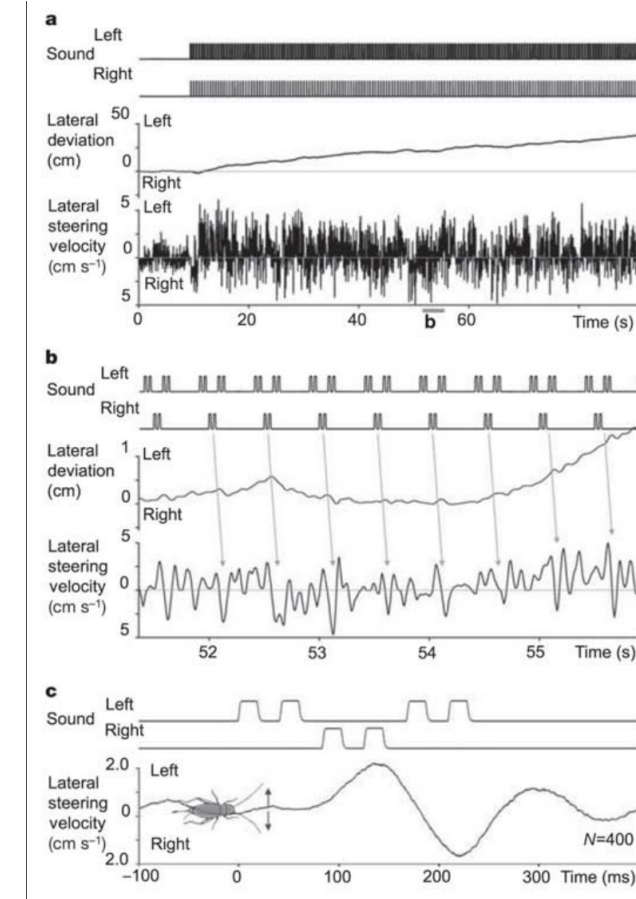
Hedwig & Poulet (2004), Fig. 1(a–b). Cricket trackball and resolved stepping.

Split songs elicit rapid steering corrections

A **split song** distributes successive parts of one chirp across two speakers. Hedwig and Poulet presented six-pulse chirps in which pairs of consecutive pulses came from opposite sides, each speaker positioned 45° from the cricket's forward axis.

Pulses lasted 21 ms and were separated by 21-ms intervals. Lateral steering followed the active side with a latency of approximately 55–60 ms and peak velocity near 5 cm/s. Each pulse pair produced a small lateral displacement of about 1 mm.

The animal's overall path favored the side presenting more pulses, yet its individual corrections followed both sides. Net deviation therefore conceals alternating stimulus-linked responses. This result challenges a purely whole-chirp account of immediate steering, but does not imply that temporal pattern recognition is absent from longer-term responsiveness.



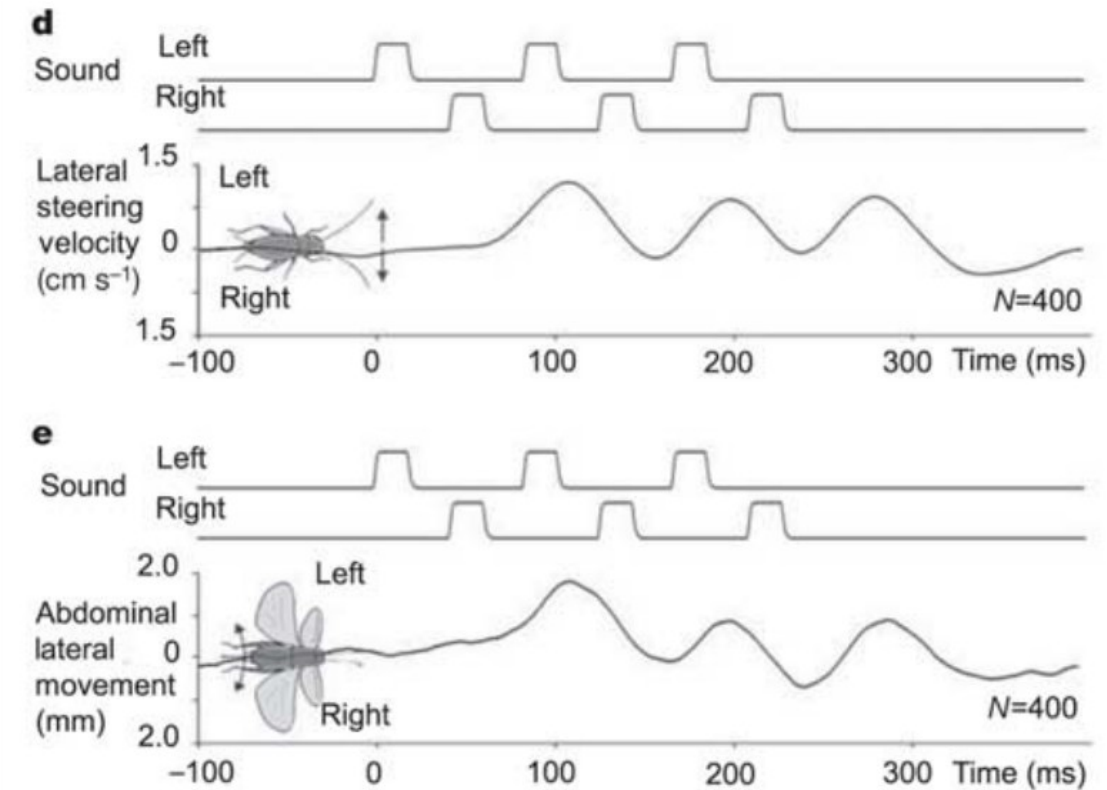
Hedwig & Poulet (2004), Fig. 2(a–c). Rapid steering toward split-song pulses.

Walking and flight retain pulse-linked steering

Auditory orientation can be expressed through different motor systems. During walking, cricket leg movements produce lateral trajectory changes; during tethered flight, abdominal movement serves as a measurable steering response toward a sound source.

Hedwig and Poulet tested the same females with consecutive pulses alternating between speakers. A constant wind stream elicited flight, and an optical recording system measured abdominal displacement. Flight steering followed the alternating sound pulses with a temporal pattern comparable to walking steering.

The correspondence supports rapid acoustic steering across locomotor modes rather than a response unique to one trackball mechanism. It does not establish identical descending circuitry or muscles in walking and flight. Similar behavioral dynamics can result from shared sensory processing that reaches different state-dependent motor pathways.



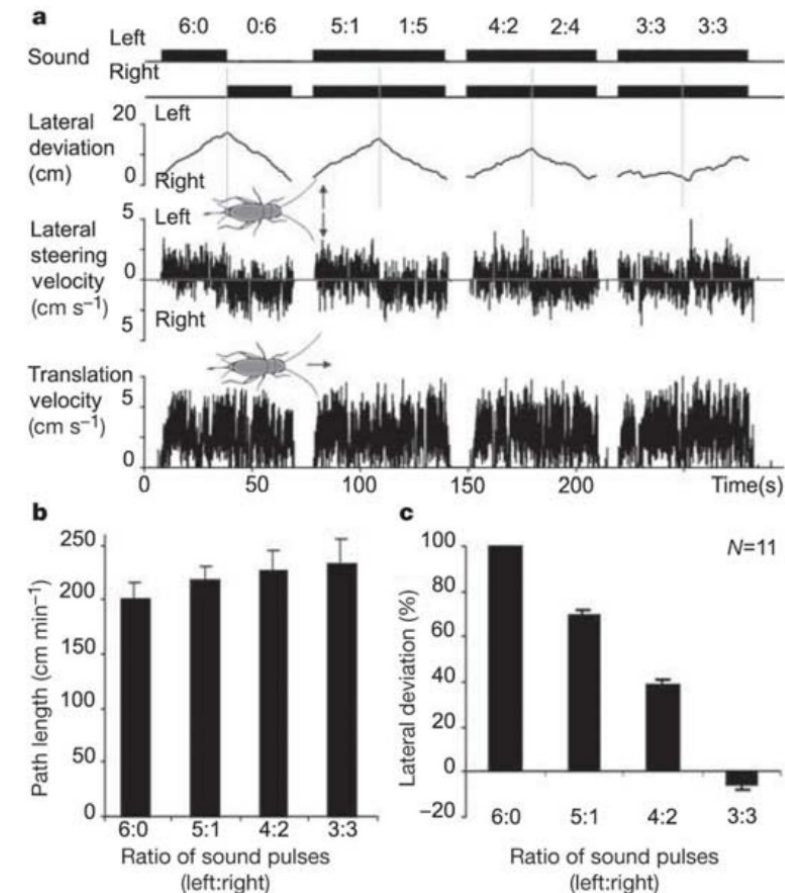
Hedwig & Poulet (2004), Fig. 2(d-e). Pulse-linked steering during walking and flight.

Pulse ratios predict net lateral deviation

When a chirp is divided between competing directions, net steering can accumulate from corrections toward individual pulses. Hedwig and Poulet varied how many of six pulses arrived from each side rather than changing the total number presented.

Females generally deviated toward the speaker providing the larger share, across ratios including 5:1, 4:2 and 3:3. At equal pulse numbers, opposing steering contributions reduced net lateral deviation even though acoustic stimulation and forward walking continued.

The result supports accumulation of rapid directional responses as an explanation for overall orientation under these split-song conditions. It does not mean the nervous system literally counts pulses with a dedicated arithmetic neuron. The measured path can emerge from unequal motor corrections without an explicit complete-chirp choice preceding every movement.



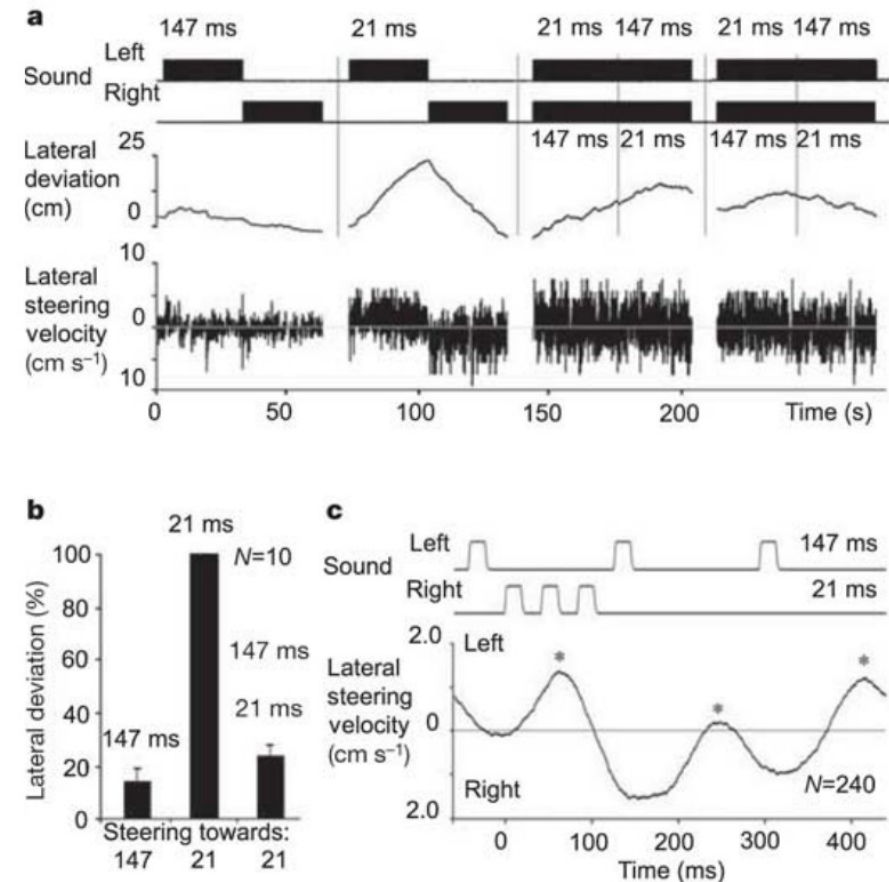
Hedwig & Poulet (2004), Fig. 3(a-c). Pulse ratios predict net lateral deviation.

Recognition modulates reactive steering

Recognition of an attractive rhythm and rapid localization of its pulses are separable processes. Hedwig and Poulet compared three-pulse patterns with 21-ms or 147-ms intervals and also presented the two patterns simultaneously from opposite sides.

The 147-ms pattern alone produced approximately 14% of the reference steering response. The 21-ms pattern was strongly attractive alone, but its net response fell to about 24% when the competing pattern was also present; the animal made corrections toward pulses from both directions.

A weakly attractive temporal pattern can therefore influence immediate steering during concurrent attractive stimulation. The authors proposed modulation of a rapid reactive pathway over a longer time course. The behavior does not identify the cellular modulation mechanism or invalidate selective brain responses measured with other temporal paradigms.



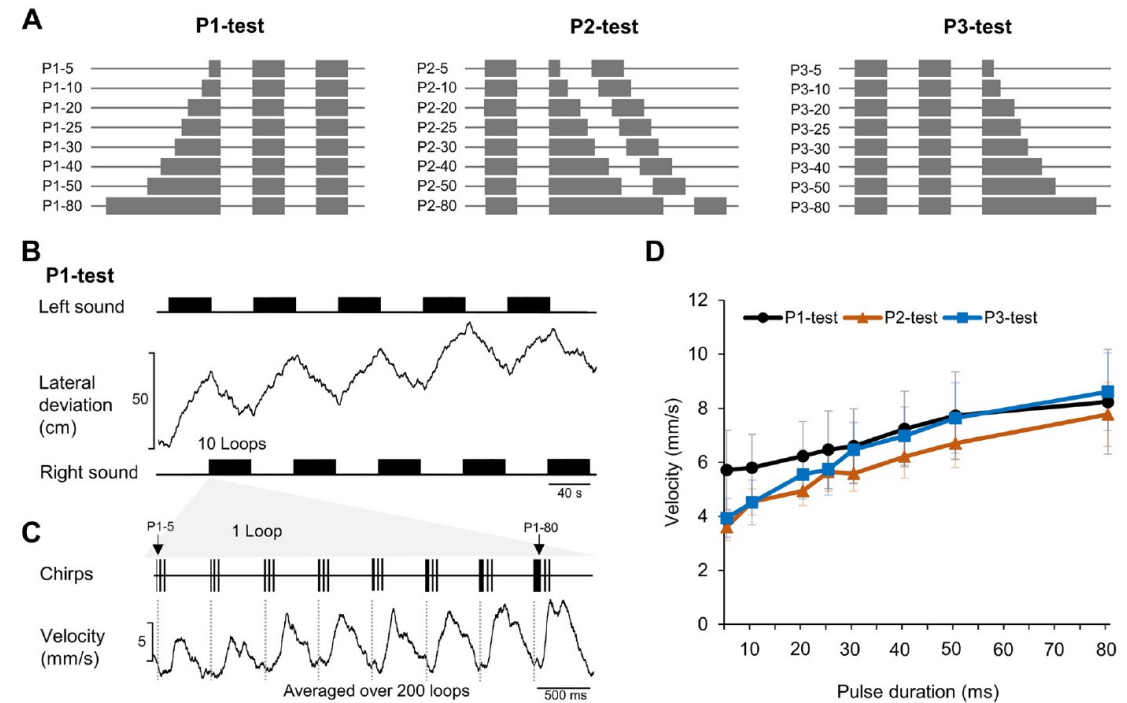
Hedwig & Poulet (2004), Fig. 4(a–c). Competing temporal patterns redirect steering.

Duration tests challenge the canonical circuit

Zhang and Hedwig varied one pulse within a three-pulse chirp while recording phonotaxis and identified brain neurons in female *Gryllus bimaculatus*. This tests timing sensitivity beyond simply changing every pulse and pause together.

The selected first, second or third pulse ranged from 5 to 80 ms. Other pulses and silent intervals remained 20 ms; chirps repeated every 500 ms with a 4.8-kHz carrier. Alternating left and right presentation reduced persistent directional bias.

Phonotactic steering increased with longer test pulses more broadly than expected from the selective output of the canonical recognition circuit. Because these test patterns were embedded in loops containing attractive chirps, the behavior concerns that stimulation context. It does not establish that isolated long-pulse songs are generally preferred or that previous period-selectivity results were incorrect.



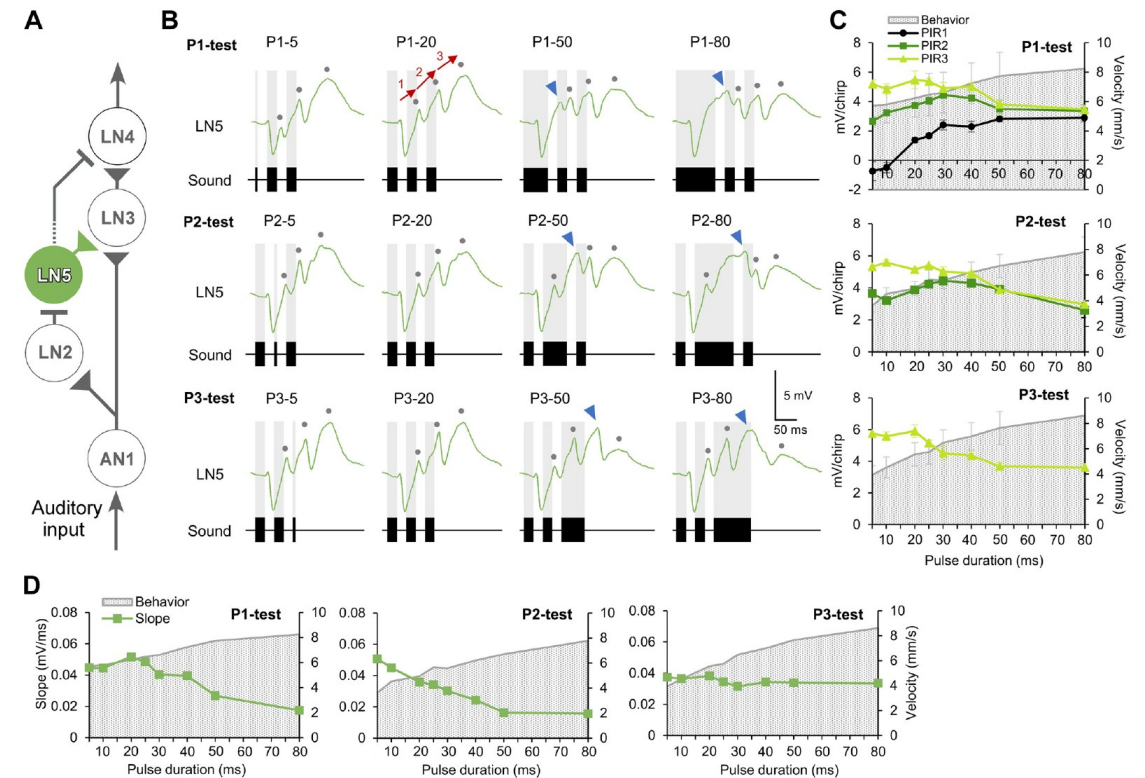
Zhang & Hedwig (2023), Fig. 1(A–D). Pulse-duration tests and phonotactic responses.

LN5 rebound depends on duration and position

Zhang and Hedwig recorded LN5 during the three pulse-position tests. The cell remained nonspiking and exhibited alternating inhibitory responses and rebound depolarizations, but the individual rebounds were not interchangeable across positions in the chirp.

Extending the first pulse from 5 toward 80 ms increased its subsequent rebound, and pulses longer than about 30 ms could produce an additional deflection before the main rebound. Comparable extensions at other pulse positions did not yield identical changes.

These responses complicate a simple fixed-delay account in which every pulse generates the same delayed signal. Intrinsic and network-dependent membrane dynamics interact with preceding sound history. The new recordings do not molecularly identify the conductance responsible, nor do they establish that LN5 alone explains the broad behavioral response to long pulses.



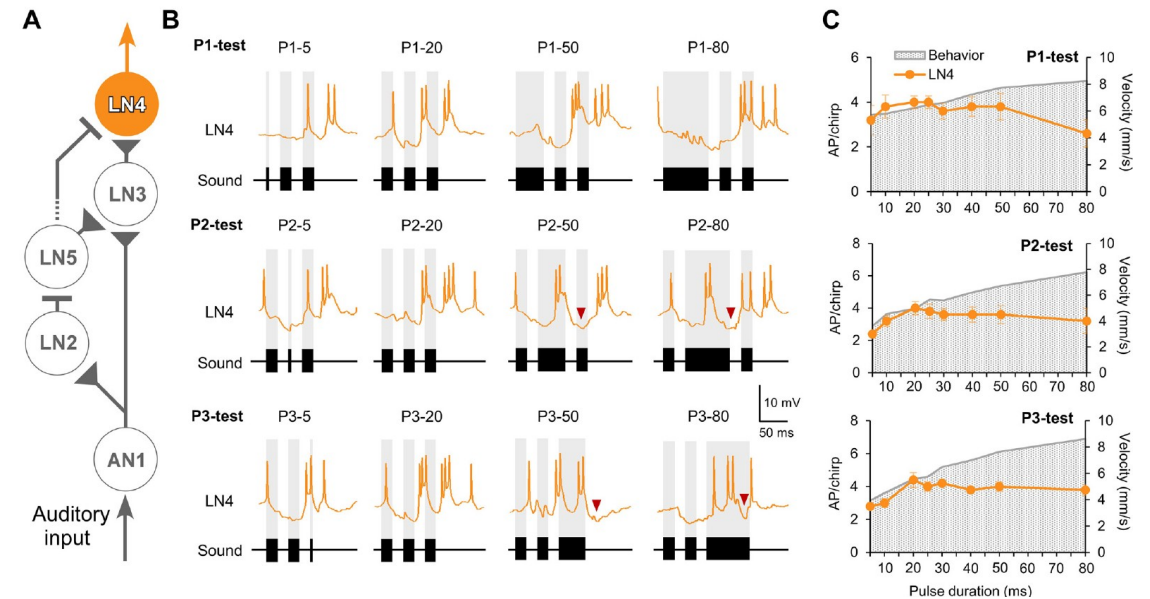
Zhang & Hedwig (2023), Fig. 3(A–D). LN5 rebound during varied pulse durations.

Long pulses recruit late inhibition in LN4

The feature-detector neuron LN4 receives excitation and inhibition rather than merely reproducing sound duration. Zhang and Hedwig examined its responses when a single pulse within a chirp was lengthened while neighboring pulse and interval values remained normal.

A long second pulse of 50 or 80 ms elicited an early excitatory burst followed by inhibition. The associated depolarization was restricted to approximately 34 ms, shorter than the ongoing sound. Long third pulses likewise produced brief excitation terminated by inhibition.

This truncation supports a duration-limiting process within neural recognition output, even when behavior remains broadly responsive. The cellular origin of the additional inhibition was not established by a targeted intervention. The response cannot be explained solely by assuming that more sound necessarily produces proportionally more LN4 spikes.



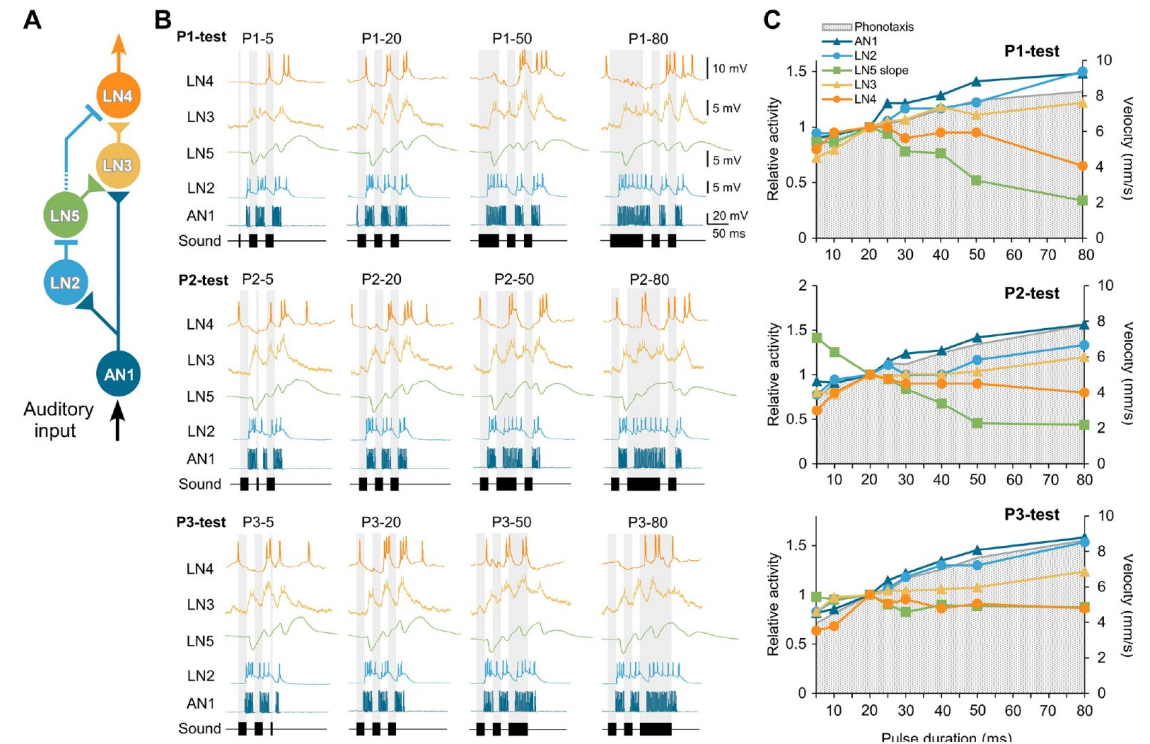
Zhang & Hedwig (2023), Fig. 6(A–C). LN4 activity and pulse-duration tuning.

The circuit does not explain every steering response

Zhang and Hedwig compared AN1 and LN2–LN5 with the phonotactic responses to the same pulse-position paradigms. The early pathway tended to increase activity with longer pulses, whereas the final selective output did not match all behavioral changes.

Across 5–80-ms test pulses, AN1 and LN2 output broadly rose with duration, while LN4 responses could decline or remain comparatively restricted. Long pulses also produced additional activity in LN3 and modified LN5 rebound, exposing computations absent from a single uniform pulse-period description.

The five-neuron circuit provides strong physiological evidence for interval recognition but is insufficient as a complete account of context-dependent steering. Additional processing, duration filtering and modulation remain plausible contributions. The study establishes the mismatch and recorded response dynamics without identifying every missing pathway or replacing the original circuit with a fully demonstrated alternative.



Zhang & Hedwig (2023), Fig. 8(A–C). Circuit responses diverge from behavioral tuning.

Key takeaways

Inherited matching. Species and hybrid song preferences support inherited sender–receiver matching, but do not identify a shared gene or channel.

Directional acoustics. Foreleg tympana combine external sound with tracheal inputs; contralateral-spiracle phase strongly contributes to forward directionality.

Input and selectivity. Thoracic and ascending cells can copy pulses without recognizing their rhythm; local brain processing creates more selective output.

Rebound and coincidence. LN5 generates intrinsic offset-linked rebound; delayed excitation combines with AN1 input in LN3, and LN4 integrates enhanced excitation with inhibition.

Recognition and steering. Selective recognition and fast directional corrections are distinct interacting processes; net phonotaxis can accumulate from steering toward individual pulses.

Evidence and limits. Proposed connections and molecular mechanisms require their own causal tests, and the five-neuron circuit does not explain all context-dependent pulse-duration responses.